Sushil Janardan

manuscript

B.Tech Tejas

Document Details

Submission ID
trn:oid:::3618:156538359

Submission Date
Oct 30, 2026, 4:31 PM GMT+5:30

Download Date
Oct 30, 2026, 4:34 PM GMT+5:30

File Name
manuscript.pdf

File Size
1.4 MB

30 Pages

17,163 Words

92,512 Characters

19% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

Match Groups

110 Not Cited or Quoted 18%
Matches with neither in-text citation nor quotation marks

21 Missing Quotations 1%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

8% Internet sources
8% Publications
16% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **110** Not Cited or Quoted 18%
  Matches with neither in-text citation nor quotation marks
- **21** Missing Quotations 1%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

8% Internet sources
8% Publications
16% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Student papers	The University of the West of Scotland on 2026-10-14 — 9%
2	Internet	arxiv.org — 1%
3	Student papers	University of Wales Swansea on 2024-05-05 — <1%
4	Internet	mdpi-res.com — <1%
5	Internet	www.mdpi.com — <1%
6	Student papers	Higher Education Commission Pakistan on 2024-11-09 — <1%
7	Student papers	University of Wollongong on 2025-03-26 — <1%
8	Internet	d-nb.info — <1%
9	Internet	ebin.pub — <1%
10	Internet	export.arxiv.org — <1%

11 Student papers
University of Nottingham on 2026-04-27 <1%

12 Student papers
University of Macau on 2014-09-21 <1%

13 Internet
osuva.uwasa.fi <1%

14 Internet
chinacpb.com <1%

15 Publication
Hongmin Gao, Yiyan Zhang, Zhonghao Chen, Chenming Li. "A Multiscale Dual-bra... <1%

16 Publication
Khan, Zirak. "Towards Efficient Deep Learning for Hyperspectral Imaging-Based F... <1%

17 Internet
research-repository.griffith.edu.au <1%

18 Publication
Guangzhe Zhao, Guangyun Liu, Leyuan Fang, Bing Tu, Pedram Ghamisi. "Multiple... <1%

19 Internet
api.research-repository.uwa.edu.au <1%

20 Internet
www.nature.com <1%

21 Publication
Dengxiang Liu, Youqiang Zhang, Jiajun Li, Bisheng Wang, Boshan Shi, Guo Cao, H... <1%

22 Publication
Buyi Zhang, Chunmei Qing, Xiangmin Xu, Jinchang Ren. "Spatial residual blocks c... <1%

23 Publication
Majid Seydgar, Amin Alizadeh Naeini, Mengmeng Zhang, Wei Li, Mehran Satari. "... <1%

24 Publication
Palvi Gupta, Himani Tyagi, Nidhi Gupta. "Learning Adaptive Gated Fusion of Conv... <1%

25 Internet
c.coek.info <1%

26 Internet
dokumen.pub <1%

27 Publication
Kim, Subin. "DBSSNet: Dual-Branch Spectral-Spatial Network With Data-Driven an... <1%

28 Publication
Peace, J. Brennan. "Robust Recognition Through Geometric and Semantic Structu... <1%

29 Student papers
University of Keele on 2026-01-11 <1%

30 Student papers
Mount Kenya University on 2020-09-20 <1%

31 Publication
Yuchao Feng, Honghui Xu, Jiawei Jiang, Hao Liu, Jianwei Zheng. "ICIF-Net: Intra-sc... <1%

32 Internet
jgit.kntu.ac.ir <1%

33 Publication
Bai, Di . "A Hyperspectral Image Classification Approach to Pigment Mapping of H... <1%

34 Publication
Gwangbin Kim, Seokhyun Hwang, Minwoo Seong, Dohyeon Yeo, Daniela Rus, Seu... <1%

35 Publication
Leiquan Wang, Yao Lin, Jinyun Liu, Zhongwei Li, Chunlei Wu. "Siamese spectral at... <1%

36 Student papers
University of Northumbria at Newcastle on 2026-08-24 <1%

37 Student papers
Khalifa University of Science Technology and Research on 2023-10-31 <1%

38 Publication
Shichao Zhang, Jiahua Zhang, Xiaopeng Wang, Jingwen Wang, Zhenjiang Wu. "ELS... <1%

39 Student papers
University of Huddersfield on 2025-07-08 <1%

40 Internet
ddescholar.acemap.info <1%

41 Student papers
Central Institute of Technology Kokrajhar on 2026-10-14 <1%

42 Publication
Li, Yijian. "Synergistic Multi-Task Learning for Skin Lesion Analysis: A Comparativ... <1%

43 Publication
Tung-tsu Chu. "The History of Chinese Feudal Society", Routledge, 2020 <1%

44 Student papers
University of Birmingham on 2021-05-07 <1%

45 Publication
Zhaohui Xue, Qi Xu, Mengxue Zhang. "Local Transformer with Spatial Partition Re... <1%

46 Publication
Zilong Zhong, Jonathan Li, Zhiming Luo, Michael Chapman. "Spectral–Spatial Resi... <1%

47 Internet
deepai.org <1%

48 Internet
dehesa.unex.es <1%

49 Internet
downloads.hindawi.com <1%

50 Publication
Dalal AL-Alimi, Zhihua Cai, Mohammed A.A. Al-qaness, Abdelghani Dahou, Eman ... <1%

51 Publication
Jiahui Qu, Zhangchun Xu, Wenqian Dong, Song Xiao, Yunsong Li, Qian Du. "A Spat... <1%

52 Publication
Ronghua Shang, Jinhong Ren, Songling Zhu, Weitong Zhang, Jie Feng, Yangyang L... <1%

53 Publication
Scheck, Kevin. "Personalizing Myoelectric Silent Speech Interfaces via Cross-Spea... <1%

54 Publication
Shaoliang Shi, Xuyang Li, Xiangsuo Fan, Qi Li. "Hyperspectral Image Classification... <1%

55 Student papers
University of Glasgow on 2026-04-24 <1%

56 Publication
Wenhui Guo, Guixun Xu, Weifeng Liu, Baodi Liu, Yanjiang Wang. "CNN-combined ... <1%

57 Publication
Xinhui Wang, Bin Yan, Pengyu Guo, Xiaolong Yang, Hongyu Liu, Lu Cao, Yong Liu, ... <1%

58 Internet
dehesa.unex.es:8080 <1%

59 Internet
personal.utdallas.edu <1%

60 Internet
publikationen.bibliothek.kit.edu <1%

61 Internet
rgu-repository.worktribe.com <1%

62 Publication
"Hyperspectral Anomaly Detection via Background Estimation and Adaptive Weig... <1%

63 Publication
Bhabani Prasanna Pattanaik, Dhirendra Kumar Shukla, Sambit Satpathy, Kamala... <1%

64 Publication
Farhan Ullah, Irfan Ullah, Khalil Khan, Salabat Khan, Farhan Amin. "Advances in d... <1%

65 Publication
Jie Feng, Licheng Jiao, Fang Liu, Tao Sun, Xiangrong Zhang. "Unsupervised featur... <1%

66 Publication
Ma, Kenneth Yeonkong. "Iterative Training Sampling and Active Learning Approa... <1%

67 Student papers
Universiti Putra Malaysia on 2018-08-11 <1%

68 Student papers
University of Alabama, Huntsville on 2020-05-30 <1%

69 Internet
coek.info <1%

70 Internet
serwiss.bib.hs-hannover.de <1%

71 Internet
uh-ir.tdl.org <1%

72 Internet
www.arxiv-vanity.com <1%

73 Publication
Priyanka Harjule, Azizur Rahman, Basant Agarwal, Vinita Tiwari. "Computational ... <1%

74 Publication
Qingming Leng, Haiou Yang, Junjun Jiang, Qi Tian. "Adaptive MultiScale Segment... <1%

75 Publication
Raj Bahadur Singh, Aloke Datta. "RMA-Net: Residual Multiscale Attention CNN for... <1%

76 Publication
Tingrui Feng, Yali Wang, Chuan Fu, Bo Du, Fulin Luo. "MMFormer: Macro–Micro Tr... <1%

77 Student papers
University of Macau on 2014-08-07 <1%

78 Publication
Bing Tu, Wangquan He, Wei He, Xianfeng Ou, Antonio J Plaza. "Hyperspectral Clas... <1%

79 Student papers
University of Surrey on 2018-08-19 <1%

80 Publication
Xu, Jun, Renlong Hang, and Qingshan Liu. "Patch-based active learning (PTAL) for ... <1%

SGSCF-Net: Spectral-Guided Spatial Cross-Attention Fusion Network for Hyperspectral Image Classification

TEJAS SURYA, National Institute of Technology (NIT) Raipur, India
AUTHOR ABC, Institute for Clarity in Documentation, USA
AUTHOR XYZ, University of Advanced Research, United Kingdom

Hyperspectral image (HSI) classification is a critical task in remote sensing Earth observation. While deep convolutional neural networks and vision transformers have advanced classification accuracy, they frequently suffer from parameter redundancy, high computational latency, and vulnerability to spectral noise caused by atmospheric absorption bands. Furthermore, conventional architectures rely predominantly on first-order spatial pooling, discarding crucial high-order co-activation dependencies between spectral channels and spatial domains. To resolve these limitations without incurring heavy transformer or recurrent state-space model overheads, we propose the **Spectral-Guided Spatial Cross-Attention Fusion Network (SGSCF-Net)**. SGSCF-Net integrates three specialized architectural modules: (1) a **Spectral Band Attention Module (SBAM)** that operates directly on raw input bands prior to convolution to suppress uninformative absorption channels; (2) a **Complementary Branch Spectral Attention (CBSA)** mechanism that enforces strict feature specialization between dual spectral-only ($1 \times 1 \times 7$) and joint spectral-spatial ($3 \times 3 \times 7$) 3D convolution streams via complementary gating; and (3) a **Second-Order Covariance Fusion Gate (SCFG)** that captures fine-grained channel-spatial co-activation through bilinear outer-product covariance pooling. To safeguard decision boundaries on heavily skewed land-cover distributions, training is governed by a minority-aware hybrid objective combining margin-scheduled Spread Loss and class-weighted Focal Loss. SGSCF-Net eliminates the excessive computational complexity of quadratic self-attention while preserving rich spectral-spatial representation capacity. Extensive experiments on three benchmark datasets (Indian Pines, Pavia University, and Salinas Valley) demonstrate that SGSCF-Net achieves a substantial reduction in model parameters and training latency while matching or exceeding the classification performance of state-of-the-art methods, maintaining robust generalization even under severe sample scarcity.

Index Terms—Hyperspectral image (HSI) classification, neural network, spectral-spatial attention, feature fusion.

CCS Concepts: • **Computing methodologies** → **Neural networks**; **Computer vision problems**; *Supervised learning by classification*; • **Applied computing** → *Environmental sciences.*

Additional Key Words and Phrases: Hyperspectral Image Classification, Spectral-Spatial Feature Extraction, Attention Mechanism, Cross-Attention Fusion, Second-Order Covariance, Deep Learning

ACM Reference Format:

Tejas Surya, Author ABC, and Author XYZ. 2018. SGSCF-Net: Spectral-Guided Spatial Cross-Attention Fusion Network for Hyperspectral Image Classification. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 30 pages. https://doi.org/XXXXXXX.XXXXXXX

Authors' Contact Information: Tejas Surya, stkisan098.btech2023@it.nitrr.ac.in, National Institute of Technology (NIT) Raipur, Raipur, Chhattisgarh, India; Author ABC, author.abc@institution.edu, Institute for Clarity in Documentation, Dublin, Ohio, USA; Author XYZ, author.xyz@university.ac.uk, University of Advanced Research, London, United Kingdom.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.
Manuscript submitted to ACM

turnitin Page 10 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

1 Introduction

Hyperspectral image (HSI) acquisition captures continuous, narrow spectral bands across hundreds of contiguous wavelengths alongside rich two-dimensional spatial structures [3, 36]. The resulting three-dimensional data cube provides a detailed diagnostic spectral signature for every spatial pixel, frequently referred to as a spectral fingerprint [12, 14]. By revealing subtle chemical and physical material properties imperceptible to standard trichromatic RGB sensors, HSI technology serves as a cornerstone in Earth observation. It is widely deployed across precision agriculture, ecological monitoring, mineral exploration, and urban surveyance.

HSI classification aims to assign a distinct land-cover semantic label to each spatial pixel by simultaneously interpreting complex spectral-spatial correlations. In recent years, deep learning models have emerged as the dominant paradigm for HSI classification [7, 52]. Specifically, convolutional neural networks (CNNs), particularly 3D CNNs and hybrid 2D-3D frameworks such as SSRN [54] and HybridSN [39], extract localized spectral-spatial features by applying learnable 3D kernel operations directly over contiguous bands, establishing a theoretical foundation for 3D volumetric convolution [2] built upon deep residual principles [20]. However, cascading deep multi-scale 3D convolutional layers [21] rapidly inflates parameter counts, precipitating substantial memory consumption and extended training latencies, while struggling to preserve complex parcel boundaries and diverse region structures across heterogeneous agricultural scenes [51].

To overcome the local receptive field constraints of CNNs, Vision Transformers (ViTs) [9] and tokenized architectures such as HSI-SSFTT [44], Spatial-Spectral Transformer [23], HiT [50], and MorphFormer [38] have been introduced to capture global spatial-spectral dependencies via self-attention. However, standard self-attention mechanisms scale with quadratic computational complexity $O(N^2)$ with respect to token sequence length. On dense HSI patches, this quadratic scaling imposes an acute computational bottleneck during both training and inference. Furthermore, hyperspectral data inherently suffer from non-uniform spectral noise, notably strong atmospheric water-vapor absorption bands. Most existing deep networks apply convolutional filters indiscriminately across all spectral bands without pre-filtering, allowing noise from uninformative bands to propagate into downstream latent representations. In addition, conventional architectures compress spatial dimensions using first-order global average pooling, discarding high-order co-activation statistics between specific spatial positions and discriminative spectral channels. Compounding these structural challenges, real-world HSI benchmarks feature severe class imbalance, where minority classes contain as few as tens of training samples, causing standard cross-entropy loss functions to experience catastrophic boundary collapse.

Rather than escalating model capacity through heavier Transformers or recurrent state-space models such as Mamba [16], an effective solution lies in targeted spectral gating and high-order covariance interaction. Unlike post-convolution attention blocks (e.g., CBAM [48]) that recalibrate only after spatial blur has propagated, applying attention directly to raw input bands before any convolution enables the network to adaptively suppress noisy atmospheric channels at the earliest processing stage. Second, in dual-branch feature extractors combining spectral-only and joint spectral-spatial convolutions, enforcing a strictly complementary gating schedule (α versus $1 - \alpha$) compels each stream to learn specialized, orthogonal representations rather than redundant features. Third, replacing first-order spatial pooling with second-order covariance pooling enables the model to capture fine-grained bilinear correlations between spatial activations and spectral channels without excessive parameter growth.

To this end, we design the **Spectral-Guided Spatial Cross-Attention Fusion Network (SGSCF-Net)**, an end-to-end framework tailored for efficient, noise-resilient hyperspectral classification. SGSCF-Net integrates three novel modules:

turnitin Page 10 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 11 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

(1) **Spectral Band Attention Module (SBAM):** Positioned at the network entry, SBAM operates directly on raw spectral channels prior to convolution. It adaptively recalibrates input channels via a spatial-squeeze MLP bottleneck, attenuating uninformative absorption bands early.
(2) **Complementary Branch Spectral Attention (CBSA):** Across the dual 3D entry streams (a $1 \times 1 \times 7$ spectral branch and a $3 \times 3 \times 7$ spatial-spectral branch), CBSA computes a channel-wise gating parameter $\alpha \in (0, 1)$ via a lightweight depthwise 1D convolution. Fusing the branches as $\alpha \odot a + (1 - \alpha) \odot b$ strictly enforces complementary feature specialization.
(3) **Second-Order Covariance Fusion Gate (SCFG):** After spectral feature compression, SCFG computes the outer product between global channel descriptors and spatial descriptors ($G = f_{\text{ch}} \otimes f_{\text{sp}}$), normalizing the resulting bilinear interaction tensor via Frobenius norm to gate spatial activations.

To safeguard decision boundaries on skewed class distributions, SGSCF-Net is optimized using a minority-aware hybrid objective combining margin-scheduled Spread Loss [41] and class-weighted Focal Loss [31], drawing upon margin optimization and sample-effective regularization principles [5, 19]. Containing approximately 308K parameters, SGSCF-Net operates with zero self-attention overhead while preserving high representational capacity.

The primary contributions of this work are summarized as follows:

(1) We introduce an input-level Spectral Band Attention Module (SBAM) that filters noisy and uninformative raw spectral bands prior to feature extraction, mitigating noise propagation throughout deeper layers.
(2) We formulate a Complementary Branch Spectral Attention (CBSA) mechanism that explicitly couples spectral-only and spatial-spectral 3D convolutions through complementary gating, ensuring optimal feature diversity without parameter inflation.
(3) We propose a Second-Order Covariance Fusion Gate (SCFG) that models bilinear channel-spatial co-activation, capturing fine-grained dependencies overlooked by standard first-order pooling operators.
(4) We conduct extensive experiments spanning more than 120 configurations across three benchmark datasets (Indian Pines, Pavia University, and Salinas Valley). Across eight standardized state-of-the-art baselines, SGSCF-Net attains the highest Average Accuracy on Indian Pines (94.82%) at 98.05% Overall Accuracy, and reaches top-tier Overall Accuracy on Pavia University (99.82%) and Salinas Valley (99.92%), while demonstrating strong robustness under severe 1% data scarcity.

The remainder of this article is organized as follows. Section II provides a review of related literature across convolutional, attention-based, and transformer architectures for HSI classification. Section III presents the formal mathematical formulation of SGSCF-Net, detailing SBAM, CBSA, SCFG, and the hybrid training objective. Section IV reports the experimental setup, comparative baseline evaluations, and computational complexity analyses. Section V discusses ablation studies and data scarcity stress testing. Finally, Section VI concludes the paper.

2 Related Work

2.1 Convolutional Neural Networks for Hyperspectral Classification

Deep learning models have substantially superseded conventional machine learning methods, such as support vector machines [33] and kernel methods [4], by eliminating manual handcrafted feature engineering. Within this paradigm, convolutional neural networks (CNNs) have received widespread adoption for hyperspectral image (HSI) feature extraction [6, 7]. Early approaches focused on autoencoder architectures and one-dimensional CNNs (1D CNNs) to capture continuous spectral profiles across spectral curves [7, 26]. While effective at distinguishing fine spectral

turnitin Page 11 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 12 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

discrepancies, 1D CNNs entirely disregard the spatial context of adjacent pixels, making them susceptible to label noise in heterogeneous land covers.

To exploit spatial correlation, two-dimensional CNNs (2D CNNs) were subsequently adopted by applying spatial convolutions on principal-component-reduced bands [52]. However, spatial-only processing discards fine diagnostic spectral signatures across contiguous absorption bands. To overcome this limitation, three-dimensional convolutional neural networks (3D CNNs) were introduced to jointly extract spectral and spatial features directly from the raw volumetric HSI cube [2]. A prominent milestone is the Spectral-Spatial Residual Network (SSRN) [54], which employs consecutive 3D residual blocks to learn joint spatial-spectral representations. Similarly, Roy *et al.* proposed HybridSN [39], integrating cascaded 3D convolutions with downstream 2D convolutions to mitigate computational complexity. Furthermore, multi-scale 3D CNNs [21] and diverse region-based CNNs [51] have explored multi-scale receptive field hierarchies to characterize spatial contextual variance across complex terrains.

Despite their success, standard 3D CNNs exhibit two critical shortcomings: (1) stacking multi-scale 3D convolutional kernels induces severe parameter explosion and quadratic memory overhead; and (2) existing CNN architectures apply kernels uniformly across all spectral channels without input-level pre-filtering, allowing strong atmospheric water-vapor noise to degrade deep representation learning.

2.2 Attention Mechanisms, Vision Transformers, and State-Space Models

To circumvent the localized receptive field constraint of standard convolution, attention mechanisms have been extensively investigated to dynamically weigh informative features. The Squeeze-and-Excitation Network (SENet) [25], Convolutional Block Attention Module (CBAM) [48], and Efficient Channel Attention (ECA-Net) [46] introduced channel and spatial attention through global average pooling and local 1D convolutions. In remote sensing, Sun *et al.* introduced the Spectral-Spatial Attention Network (SSAN) [43], which decouples attention into spatial and spectral branches. Subsequent architectures incorporated attention-aided CNNs [18], residual attention mechanisms [55], and connected feature-grouped attention [17]. In parallel, reinforcement learning and attention strategies have been explored for explicit band selection and spectral channel weighting [34]. Furthermore, dual-branch frameworks such as the Attention-Based Adaptive Spectral-Spatial Kernel ResNet (A2S2K-ResNet) [40], Double-Branch Dual-Attention Networks [29], and CSCANet [45] utilize multi-branch streams to extract complementary spatial-spectral features. However, existing dual-branch frameworks frequently compute a single shared scalar weight or combine features via simple summation, failing to enforce feature specialization between the complementary streams.

Concurrently, Vision Transformers (ViTs) [9] have redefined computer vision by modeling long-range contextual relationships through multi-head self-attention. In hyperspectral remote sensing, architectures such as HSI-SSFTT [44], SpectralFormer [24], Spatial-Spectral Transformer [23], HiT [50], MorphFormer [38], and improved Transformer variants [37] have achieved high classification accuracy by tokenizing spatial-spectral patches. Recently, state-space models such as Mamba [16] and MambaHSI [30] have explored linear-complexity selective scan mechanisms. Nevertheless, self-attention mechanisms fundamentally incur quadratic computational complexity $O(N^2)$ with respect to spatial-spectral token sequence length, leading to severe computational latency and hardware memory bottlenecks on dense HSI patches.

2.3 Second-Order Covariance Representations and Class Imbalance

A prominent limitation across conventional CNNs and Transformer backbones is their reliance on first-order statistics. Standard architectures compress spatial feature maps using Global Average Pooling (GAP), which calculates the

turnitin Page 12 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 13 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

arithmetic mean of spatial activations. While computationally trivial, GAP implicitly assumes spatial uniformity and discards crucial second-order co-activation statistics between spatial positions and spectral channels. In fine-grained computer vision, Bilinear CNNs [32] and global second-order pooling networks [13, 28] demonstrated that second-order outer-product covariance pooling captures rich co-occurrence statistics. In HSI classification, local covariance matrix representations [10], multiscale covariance maps [22], attention-based second-order pooling [49], and mixed convolution covariance pooling [53] confirmed the potency of covariance representations for capturing spatial-spectral correlations. Furthermore, capsule networks [35] and Deep Matrix Capsules (DMC) with EM routing [47] incorporated high-order matrix poses to model spatial hierarchies, demonstrating high accuracy but requiring specialized routing iterations.

Compounding these architectural bottlenecks, hyperspectral benchmark datasets suffer from severe class imbalance. Certain agricultural categories (e.g., Oats in Indian Pines) possess as few as tens of samples, whereas dominant classes comprise thousands of samples. Standard cross-entropy loss gradients become heavily dominated by prevalent background classes, precipitating catastrophic boundary misclassifications on minority land covers. While Focal Loss [31] addresses hard-example mining through dynamic modulating factors, effective sample volume weighting [8] and margin-based losses such as Spread Loss [41] and LDAM [5] enforce explicit margins between ground-truth activations and competing categories. In addition, minority oversampling techniques such as HyperGAMO [19] and feature-rotation ensembles [42] emphasize the necessity of safeguarding decision boundaries under extreme sample scarcity.

To reconcile these divergent challenges, **SGSCF-Net** introduces an integrated solution: (1) an input-level Spectral Band Attention Module (SBAM) that filters atmospheric absorption noise prior to convolution; (2) a Complementary Branch Spectral Attention (CBSA) mechanism that enforces orthogonal branch specialization through complementary gating (α vs $1-\alpha$); (3) a Second-Order Covariance Fusion Gate (SCFG) that models bilinear channel-spatial co-activation; and (4) a minority-aware hybrid objective combining margin-scheduled Spread Loss and Focal Loss. This yields state-of-the-art representation capacity with only ~308K parameters and zero quadratic self-attention latency.

3 Methodology

This section provides a formal mathematical formulation of the proposed Spectral-Guided Spatial Cross-Attention Fusion Network (SGSCF-Net). We first establish the problem formulation and high-level architectural pipeline for hyperspectral image classification. Subsequently, we detail each constituent module in sequence: the Spectral Band Attention Module (SBAM) for input-level spectral recalibration, the Complementary Branch Spectral Attention (CBSA) for orthogonal dual-branch fusion, the Second-Order Covariance Fusion Gate (SCFG) for bilinear channel-spatial interaction, and the minority-aware hybrid training objective.

3.1 Overall Pipeline and Problem Formulation

Let $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ denote the raw hyperspectral data cube, where H, W, and C represent the spatial height, width, and number of contiguous spectral bands, respectively. Let $\mathcal{Y} \in \{1, 2, \ldots, C_{\text{cls}}\}^{H\times W}$ denote the corresponding ground-truth land-cover semantic label map. For any labeled target pixel located at spatial coordinate (i, j), we extract a localized spatial neighborhood cube of size $P \times P$ across all C spectral channels. To handle boundary pixels, symmetric spatial reflection padding of size $p = \lfloor P/2 \rfloor$ is applied. Each input sample is structured as a five-dimensional volumetric tensor:

$$X \in \mathbb{R}^{B\times 1\times P\times P\times C} \tag{1}$$

where B denotes the mini-batch size, and the unitary second dimension represents the input channel axis required for 3D convolutional operations.

turnitin Page 13 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 14 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

The computational dataflow of SGSCF-Net proceeds through nine structured stages:

(1) X enters the **Spectral Band Attention Module (SBAM)**, generating an adaptively recalibrated volumetric tensor $\tilde{X} \in \mathbb{R}^{B\times 1\times P\times P\times C}$ that attenuates noisy or corrupted atmospheric bands before convolutional processing.

(2) $\tilde{X}$ is fed into a **Dual 3D Entry Stream**, producing two parallel feature maps: a spectral-only stream $a \in \mathbb{R}^{B\times K\times P\times P\times C_2}$ via $1\times 1\times 7$ 3D convolution, and a joint spatial-spectral stream $b \in \mathbb{R}^{B\times K\times P\times P\times C_2}$ via $3\times 3\times 7$ 3D convolution, where $K = 16$ filters and spectral stride downsampling yield $C_2 = \lfloor (C-7)/2 \rfloor + 1$.

(3) The streams (a, b) are coupled by the **Complementary Branch Spectral Attention (CBSA)** mechanism, which computes a depthwise 1D convolutional gate $\alpha \in \mathbb{R}^{B\times K\times 1\times 1\times C_2}$ and executes complementary fusion $\alpha \odot a + (1-\alpha) \odot b$, forcing orthogonal feature specialization.

(4) The fused tensor passes through two cascaded **3D Spectral Residual Blocks** $(\mathcal{R}^{(1)}_{\text{spec}}, \mathcal{R}^{(2)}_{\text{spec}})$ equipped with 3D Efficient Channel Attention (ECA-3D) to deepen spectral representations.

(5) An adaptive 3D average pooling layer collapses the spectral dimension $C_2 \to 1$, and a $1\times 1\times 1$ 3D convolution expands the channel dimension to $C_{\text{mid}} = 128$. Squeezing the singleton spectral axis yields a 2D spatial feature map $x \in \mathbb{R}^{B\times 128\times P\times P}$.

(6) x enters the **Second-Order Covariance Fusion Gate (SCFG)**, which models bilinear outer-product interaction between channel and spatial descriptors ($G = f_{\text{ch}} \otimes f_{\text{sp}} \in \mathbb{R}^{B\times 128\times N}$), normalizes the resulting matrix via Frobenius norm, and applies a linear projection gate to recalibrate spatial activations.

(7) The recalibrated tensor is permuted and treated as a volumetric tensor $v_{\text{spat}} \in \mathbb{R}^{B\times 1\times P\times P\times 128}$ for a spatial 3D entry convolution ($3\times 3\times 128$), projecting channels back to $K = 16$, followed by two **3D Spatial Residual Blocks** $(\mathcal{R}^{(1)}_{\text{spat}}, \mathcal{R}^{(2)}_{\text{spat}})$.

(8) Global 3D average pooling compresses spatial activations into a latent feature vector $z \in \mathbb{R}^{B\times K}$, mapped to class logits $\hat{y} \in \mathbb{R}^{B\times C_{\text{cls}}}$ by a linear classification head.

(9) The complete network is trained end-to-end via a **Minority-Aware Hybrid Loss** $\mathcal{L}_{\text{Total}} = 0.7\mathcal{L}_{\text{Spread}} + 0.3\mathcal{L}_{\text{Focal}}$ with dynamic margin scheduling and inverse-frequency class weights.

3.2 Spectral Band Attention Module (SBAM)

Hyperspectral sensors capture hundreds of contiguous spectral bands, but atmospheric water vapor absorption bands (such as those around 1.4 μm and 1.9 μm) and low signal-to-noise ratios at spectral boundaries corrupt raw reflectance signatures. Conventional deep learning architectures apply convolutional kernels indiscriminately across all raw bands, permitting noisy band responses to propagate into deep feature representations. While the Squeeze-and-Excitation Network (SENet) [25] and Convolutional Block Attention Module (CBAM) [48] introduced channel and spatial attention, they operate strictly on deep latent feature maps following multiple convolutional stages. Unlike reinforcement learning band selection policies [34] or connected grouping heuristics [17] that require expensive outer optimization loops, SBAM embeds direct channel gating into the end-to-end differentiable graph.

To eliminate noise at the earliest computational stage, SBAM performs adaptive channel recalibration directly on raw spectral bands before any convolutional feature extraction. Given the input volumetric patch $X \in \mathbb{R}^{B\times 1\times P\times P\times C}$, SBAM first compresses the spatial dimensions $P\times P$ via adaptive 3D average pooling to extract a global spectral descriptor $z \in \mathbb{R}^{B\times C}$:

$$z_c = \frac{1}{P^2} \sum_{u=1}^{P} \sum_{v=1}^{P} X(b, 1, u, v, c), \quad c \in \{1, 2, \ldots, C\} \tag{2}$$

turnitin Page 14 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 15 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

To model inter-band dependencies while preserving low parameter complexity, the summary vector z is routed through a multi-layer perceptron (MLP) bottleneck consisting of two linear projections with a dimensionality reduction ratio $r = 8$:

$$w = \sigma\left(W_2 \cdot \delta(W_1 z + b_1) + b_2\right) \tag{3}$$

where $W_1 \in \mathbb{R}^{C_{\text{mid}} \times C}$, $b_1 \in \mathbb{R}^{C_{\text{mid}}}$, $W_2 \in \mathbb{R}^{C \times C_{\text{mid}}}$, and $b_2 \in \mathbb{R}^{C}$. The bottleneck channel capacity is defined as $C_{\text{mid}} = \max(\lfloor C/r \rfloor, 8)$ to prevent over-compression on datasets with fewer spectral channels. Here, $\delta(\cdot)$ denotes the rectified linear unit (ReLU) activation, and $\sigma(\cdot)$ is the standard sigmoid gating function, yielding normalized band importance weights $w \in (0,1)^{B \times C}$.

Finally, the input tensor X is recalibrated via channel-wise broadcast multiplication:

$$\tilde{X} = X \odot \mathcal{E}(w), \quad \tilde{X} \in \mathbb{R}^{B \times 1 \times P \times P \times C} \tag{4}$$

where $\mathcal{E}(w) \in \mathbb{R}^{B \times 1 \times 1 \times 1 \times C}$ denotes the broadcast expansion of w across spatial dimensions, and $\odot$ represents element-wise Hadamard multiplication. By dynamically suppressing uninformative or corrupted spectral bands before convolution, SBAM shields downstream representation learning from atmospheric noise propagation.

3.3 Complementary Branch Spectral Attention (CBSA)

Following SBAM recalibration, the refined tensor $\tilde{X}$ enters a dual-stream 3D convolutional module designed to capture multi-scale spectral and spatial structures concurrently:

$$a = \text{ReLU}\left(\text{BN}\left(\text{Conv3D}_{(1\times1\times7)}(\tilde{X})\right)\right) \in \mathbb{R}^{B \times K \times P \times P \times C_2} \tag{5}$$

$$b = \text{ReLU}\left(\text{BN}\left(\text{Conv3D}_{(3\times3\times7)}(\tilde{X})\right)\right) \in \mathbb{R}^{B \times K \times P \times P \times C_2} \tag{6}$$

where $K = 16$ denotes intermediate feature channels, and both 3D convolutions employ a spectral stride of 2, reducing the spectral dimension to $C_2 = \lfloor (C-7)/2 \rfloor + 1$. Branch a uses a $1 \times 1 \times 7$ kernel focusing strictly on pure spectral inter-band transitions without spatial expansion. Branch b applies a $3 \times 3 \times 7$ kernel with spatial padding $(1, 1, 0)$ to capture joint spatial-spectral context.

In multi-branch architectures such as A2S2K-ResNet [40], Double-Branch networks [29], and CSCANet [45], parallel streams are combined either by direct summation or by computing a single shared squeeze-and-excitation attention gate. However, assigning an identical gating scalar to both streams causes them to learn co-dependent, redundant feature representations, negating the benefit of multi-scale branching.

To enforce orthogonal feature specialization, CBSA introduces a complementary gating mechanism. First, spatial context is aggregated across the fused representations $U = a + b$:

$$U_{\text{avg}} = \frac{1}{P^2} \sum_{u=1}^{P} \sum_{v=1}^{P} (a+b)[:, :, u, v, :] \in \mathbb{R}^{B \times K \times C_2} \tag{7}$$

Next, a lightweight depthwise 1D convolution with kernel size $k_s = 3$ (grouped across K channels, requiring only $K \times k_s = 48$ trainable parameters) models localized cross-band interactions:

$$\alpha = \sigma\left(\text{Conv1D}_{k_s}(U_{\text{avg}})\right) \in \mathbb{R}^{B \times K \times C_2} \tag{8}$$

Expanding α to $\mathbb{R}^{B \times K \times 1 \times 1 \times C_2}$, the two branches are fused via complementary convex combination followed by 3D batch normalization:

$$v = \text{BN}\left(\alpha \odot a + (1 - \alpha) \odot b\right) \in \mathbb{R}^{B \times K \times P \times P \times C_2} \tag{9}$$

turnitin Page 15 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 16 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Because the gating coefficients strictly satisfy $\alpha + (1 - \alpha) \equiv 1$ for every channel and band, CBSA forces competitive resource allocation between the streams. When spectral fidelity demands sharp discrimination, $\alpha \to 1$ promotes the pure spectral branch a while suppressing spatial blurring from branch b. Conversely, in homogeneous spatial regions where neighborhood context dominates, $\alpha \to 0$ elevates branch b.

The fused representation v is deepened through two consecutive 3D residual blocks equipped with 3D Efficient Channel Attention (ECA-3D) [46]:

$$v = \mathcal{R}^{(2)}_{\text{spec}}\left(\mathcal{R}^{(1)}_{\text{spec}}(v)\right) \in \mathbb{R}^{B\times K\times P\times P\times C_2} \tag{10}$$

where each spectral residual block utilizes $1 \times 1 \times 7$ 3D convolutions with padding $(0, 0, 3)$ to preserve spatial-spectral dimensions while deepening feature abstraction.

3.4 Second-Order Covariance Fusion Gate (SCFG)

To transition from spectral-dominant processing to spatial context extraction, the spectral dimension C_2 is collapsed using adaptive 3D average pooling:

$$v_{\text{pool}} = \text{AdaptiveAvgPool3D}_{(P,P,1)}(v) \in \mathbb{R}^{B\times K\times P\times P\times 1} \tag{11}$$

A $1 \times 1 \times 1$ 3D convolution expands the channel dimension from $K = 16$ to $C_{\text{mid}} = 128$:

$$v_{\text{exp}} = \text{ReLU}\left(\text{BN}\left(\text{Conv3D}_{(1\times 1\times 1)}(v_{\text{pool}})\right)\right) \in \mathbb{R}^{B\times C_{\text{mid}}\times P\times P\times 1} \tag{12}$$

Squeezing the singleton spectral dimension yields the 2D spatial feature map $x \in \mathbb{R}^{B\times C_{\text{mid}}\times P\times P}$.

Standard spatial attention mechanisms and global pooling operators compute only first-order spatial means $\mu_c = \frac{1}{HW}\sum_i x_{c,i}$, discarding spatial co-activation structures. In heterogeneous land covers, specific semantic channels correlate with distinct spatial locations (e.g., edges versus interior textures). SCFG captures these second-order correlations through bilinear outer-product covariance modeling [32].

Let $N = P \times P$ denote the total number of spatial pixels in the patch. We derive the global channel descriptor $f_{\text{ch}} \in \mathbb{R}^{B\times C_{\text{mid}}}$ and spatial position descriptor $f_{\text{sp}} \in \mathbb{R}^{B\times N}$:

$$f_{\text{ch}}(b, c) = \frac{1}{N}\sum_{n=1}^{N} x(b, c, n), \quad f_{\text{sp}}(b, n) = \frac{1}{C_{\text{mid}}}\sum_{c=1}^{C_{\text{mid}}} x(b, c, n) \tag{13}$$

The second-order spatial-channel covariance interaction matrix $G \in \mathbb{R}^{B\times C_{\text{mid}}\times N}$ is computed via batch outer product:

$$G = f_{\text{ch}} \otimes f_{\text{sp}} = f_{\text{ch}} f_{\text{sp}}^T, \quad G(b, c, n) = f_{\text{ch}}(b, c) \cdot f_{\text{sp}}(b, n) \tag{14}$$

Each matrix entry $G(b, c, n)$ measures the bilinear co-occurrence strength between channel c and spatial coordinate n, extending second-order covariance principles from fine-grained recognition [13, 32] and HSI covariance modeling [10, 22, 49, 53] into an efficient spatial-channel gating operator without requiring iterative matrix square-root decompositions [28].

To guarantee numerical stability and invariant gradient scaling across varied patch dynamics, G is normalized by its Frobenius norm:

$$\tilde{G} = \frac{G}{\|G\|_F + \epsilon} = \frac{G}{\sqrt{\sum_{c=1}^{C_{\text{mid}}}\sum_{n=1}^{N} G(b, c, n)^2} + \epsilon} \tag{15}$$

turnitin Page 16 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 17 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

where $\epsilon = 10^{-8}$. A learnable linear projection $W_G \in \mathbb{R}^{1\times N}$ compresses the spatial interaction dimension of $\tilde{G}$ into a channel-wise modulation gate:

$$g = \sigma\left(W_G \tilde{G}\right) \in \mathbb{R}^{B\times C_{\text{mid}}} \tag{16}$$

The recalibrated feature map is obtained by element-wise modulation:

$$x_{\text{scfg}} = x \odot g \tag{17}$$

where g is broadcast across the spatial grid $P \times P$.

To process spatial context, x_{scfg} is reshaped by permuting spatial and channel axes into a 3D volumetric tensor $v_{\text{spat}} \in \mathbb{R}^{B\times 1\times P\times P\times C_{\text{mid}}}$, treating the 128 channels as the spectral/depth dimension for 3D convolution. A spatial entry convolution $\text{Conv3D}_{(3\times3\times128)}$ projects the tensor to $K = 16$ feature maps:

$$s = \text{ReLU}\left(\text{BN}\left(\text{Conv3D}_{(3\times3\times128)}(v_{\text{spat}})\right)\right) \in \mathbb{R}^{B\times K\times P\times P\times 1} \tag{18}$$

The tensor is processed through two cascaded spatial 3D residual blocks with $3 \times 3 \times 1$ convolutions and ECA-3D attention:

$$s_{\text{out}} = \mathcal{R}^{(2)}_{\text{spat}}\left(\mathcal{R}^{(1)}_{\text{spat}}(s)\right) \in \mathbb{R}^{B\times K\times P\times P\times 1} \tag{19}$$

Finally, 3D adaptive average pooling collapses spatial dimensions to extract a K-dimensional latent representation $z \in \mathbb{R}^{B\times K}$:

$$z = \text{AdaptiveAvgPool3D}_{(1,1,1)}(s_{\text{out}}) \tag{20}$$

The latent vector z is subsequently projected to class logits $\hat{y} \in \mathbb{R}^{B\times C_{\text{cls}}}$ by a linear classification head:

$$\hat{y} = W_{\text{fc}} z + b_{\text{fc}} \tag{21}$$

3.5 Minority-Aware Hybrid Loss Formulation

Hyperspectral benchmark datasets exhibit severe class imbalance. For example, in Indian Pines, dominant classes such as Soybean-mintill contain 2,468 labeled pixels, whereas rare categories such as Oats and Grass-pasture-mowed contain only 20 and 28 samples, respectively. Standard cross-entropy loss gradients become heavily dominated by prevalent categories, causing catastrophic decision boundary collapse on minority land covers.

To establish wide, robust decision boundaries while prioritizing under-represented categories, SGSCF-Net is trained with a hybrid objective combining margin-scheduled Spread Loss [41] and inverse-frequency class-weighted Focal Loss [31], drawing upon margin optimization and sample-effective regularization principles [5, 8, 19].

3.5.1 Dynamic Margin-Scheduled Spread Loss. Spread Loss directly penalizes predictions where non-target class activations fall within a specified margin m of the true class logit:

$$\mathcal{L}_{\text{Spread}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{j\neq y_i}\left(\max\left(0, m(r) - (a_{y_i} - a_j)\right)\right)^2 \tag{22}$$

where a_k represents the output activation for class k, and $m(r)$ is a dynamic margin scheduled linearly according to the training progression ratio $r = \frac{t}{T_{\text{total}}} \in [0, 1]$:

$$m(r) = m_{\min} + (m_{\max} - m_{\min}) \cdot r \tag{23}$$

turnitin Page 17 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 18 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

with $m_{\min} = 0.2$ and $m_{\max} = 0.9$. In early epochs ($r \to 0$), a relaxed margin $m = 0.2$ prevents over-penalizing uncalibrated initial weights; in late epochs ($r \to 1$), the margin expands to 0.9, compelling the model to enforce wide decision boundaries between competing categories.

3.5.2 Inverse-Frequency Class-Weighted Focal Loss. To prevent dominant majority classes from overwhelming the gradient signal and to focus optimization on hard samples, we integrate Focal Loss:

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{B}\sum_{i=1}^{B} \alpha_{y_i}(1 - p_{t,i})^{\gamma} \log(p_{t,i}) \tag{24}$$

where $p_{t,i} = \frac{\exp(\hat{y}_{i,y_i})}{\sum_k \exp(\hat{y}_{i,k})}$ is the softmax probability of the ground-truth class, $\gamma = 2.0$ is the focusing parameter, and α_c is the normalized inverse-frequency weight for class $c \in \{1, \ldots, C_{\text{cls}}\}$:

$$w_c = \frac{1}{N_c + \epsilon}, \quad \alpha_c = \frac{w_c}{\sum_{k=1}^{C_{\text{cls}}} w_k} \cdot C_{\text{cls}} \tag{25}$$

where N_c is the total count of training samples belonging to class c.

3.5.3 Combined Optimization Objective. The total end-to-end training loss is defined as:

$$\mathcal{L}_{\text{Total}} = 0.7\mathcal{L}_{\text{Spread}} + 0.3\mathcal{L}_{\text{Focal}} \tag{26}$$

This hybrid objective balances margin spread across competing categories with hard-sample focal mining, ensuring balanced gradient contributions and preventing minority class collapse under extreme sample scarcity.

4 Experiments and Comparative Analysis

This section presents an empirical evaluation of SGSCF-Net alongside eight standardized baseline models. We first describe the benchmark hyperspectral datasets and provide comprehensive category distributions. Next, we outline the experimental setup, comparative models, and evaluation metrics. Finally, we report quantitative comparisons on Indian Pines, examine computational latency, and analyze parameter efficiency.

4.1 Dataset Description

To evaluate model performance across heterogeneous spatial resolutions, sensor configurations, and land-cover categories, experiments are conducted on three standard benchmark datasets: Indian Pines, University of Pavia, and Salinas Valley. In all experiments, labeled samples are partitioned into training, validation, and testing sets following a standard 10%, 10%, and 80% stratified split per class. A structured summary of sensor properties, spatial dimensions, spectral channels, and sample allocations for each benchmark is provided in Table 1. For each benchmark, the corresponding subsection below first describes the acquisition protocol and land-cover composition, and then reports the per-class ground-truth colour palette together with the stratified sample allocation (Tables 2, 3, and 4). These palettes are retained consistently throughout the classification maps rendered for each dataset in the subsequent sections.

4.1.1 Indian Pines. The Indian Pines dataset was gathered in 1992 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [15] over an agricultural test site in Northwest Indiana. The spatial scene measures 145×145 pixels with a moderate spatial resolution of 20 m per pixel. The sensor records 224 contiguous spectral bands spanning the 0.4–2.5 μm wavelength range. Following standard experimental protocol, 20 water-absorption and noise-corrupted bands (bands 104–108, 150–163, and 220) are excluded, leaving 200 clean spectral channels. The ground truth contains

turnitin Page 18 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 19 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 1. Benchmark Hyperspectral Dataset Characteristics and Sample Allocations (10% Train / 10% Val / 80% Test Split).

Dataset	Sensor	Range (μm)	Bands	Spatial Size	Res. (m)	Classes	Labeled Pixels	Train / Val / Test
Indian Pines	AVIRIS	0.40–2.50	200	145 × 145	20.0	16	10,249	1,025 / 1,025 / 8,199
University of Pavia	ROSIS-03	0.43–0.86	103	610 × 340	1.3	9	42,776	4,277 / 4,277 / 34,222
Salinas Valley	AVIRIS	0.40–2.50	204	512 × 217	3.7	16	54,129	5,413 / 5,413 / 43,303

16 land-cover categories encompassing two-thirds agricultural crops (e.g., corn, soybeans, wheat) and one-third natural vegetation (e.g., woods, grass). A total of 10,249 pixels are labeled. Indian Pines presents acute class imbalance: dominant categories such as Soybean-mintill comprise 2,468 samples, whereas minority classes such as Oats and Grass-pasture-mowed contain only 20 and 28 samples, respectively. The per-class ground-truth colour palette and stratified sample allocation are listed in Table 2.

Table 2. Category Information of the Indian Pines Dataset. Train/Val/Test counts follow a stratified 10%/10%/80% split per class.

ID	Colour	Class Name	Train	Val	Test	Total
1		Alfalfa	5	5	36	46
2		Corn-notill	143	143	1,142	1,428
3		Corn-mintill	83	83	664	830
4		Corn	24	24	189	237
5		Grass-pasture	48	48	387	483
6		Grass-trees	73	73	584	730
7		Grass-pasture-mowed	3	3	22	28
8		Hay-windrowed	48	48	382	478
9		Oats	2	2	16	20
10		Soybean-notill	97	97	778	972
11		Soybean-mintill	246	246	1,963	2,455
12		Soybean-clean	59	59	475	593
13		Wheat	21	21	163	205
14		Woods	127	127	1,011	1,265
15		Buildings-Grass-Trees-Drives	39	39	308	386
16		Stone-Steel-Towers	9	9	75	93
		Total	**1,025**	**1,025**	**8,199**	**10,249**

4.1.2 University of Pavia. The University of Pavia dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor during an aerial flight campaign over Pavia, Northern Italy [11]. The image dimensions are 610 × 340 pixels with a fine spatial resolution of 1.3 m per pixel. The sensor covers the spectral range from 430 to 860 nm across 115 bands. After discarding 12 heavily corrupted atmospheric noise bands, 103 calibrated spectral channels are retained. The ground-truth reference map contains nine urban land-cover classes, including asphalt roads, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows. The scene contains 42,776 labeled pixels, characterized by complex geometric boundaries and high intra-class spatial variance. The per-class ground-truth colour palette and stratified sample allocation are listed in Table 3.

4.1.3 Salinas Valley. The Salinas Valley dataset was captured by the AVIRIS sensor [15] over the fertile Salinas Valley agricultural region in California. The dataset covers 512 × 217 pixels with a spatial resolution of 3.7 m per pixel. The spectrometer records 224 bands across 0.4–2.5 μm. Discarding 20 water-vapor absorption bands (bands 108–112, 154–167, and 224) yields 204 usable spectral channels. The ground truth comprises 16 agricultural land-cover classes, including multiple cultivars of lettuce at distinct growth stages (e.g., 4-week, 5-week, 6-week, and 7-week Romaine), broccoli,

turnitin Page 19 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 20 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 3. Category Information of the University of Pavia Dataset. Train/Val/Test counts follow a stratified 10%/10%/80% split per class.

ID	Colour	Class Name	Train	Val	Test	Total
1		Asphalt	663	663	5,305	6,631
2		Meadows	1,865	1,865	14,919	18,649
3		Gravel	210	210	1,679	2,099
4		Trees	306	306	2,452	3,064
5		Painted metal sheets	134	134	1,077	1,345
6		Bare Soil	503	503	4,023	5,029
7		Bitumen	133	133	1,064	1,330
8		Self-Blocking Bricks	368	368	2,946	3,682
9		Shadows	95	95	757	947
		Total	**4,277**	**4,277**	**34,222**	**42,776**

celery, grapes, and vineyard fields. A total of 54,129 pixels are labeled. The per-class ground-truth colour palette and stratified sample allocation are listed in Table 4.

Table 4. Category Information of the Salinas Valley Dataset. Train/Val/Test counts follow a stratified 10%/10%/80% split per class.

ID	Colour	Class Name	Train	Val	Test	Total
1		Brocoli_green_weeds_1	201	201	1,607	2,009
2		Brocoli_green_weeds_2	373	373	2,980	3,726
3		Fallow	198	198	1,580	1,976
4		Fallow_rough_plow	139	139	1,116	1,394
5		Fallow_smooth	268	268	2,142	2,678
6		Stubble	396	396	3,167	3,959
7		Celery	358	358	2,863	3,579
8		Grapes_untrained	1,127	1,127	9,017	11,271
9		Soil_vinyard_develop	620	620	4,963	6,203
10		Corn_senesced_green_weeds	328	328	2,622	3,278
11		Lettuce_romaine_4wk	107	107	854	1,068
12		Lettuce_romaine_5wk	193	193	1,541	1,927
13		Lettuce_romaine_6wk	92	92	732	916
14		Lettuce_romaine_7wk	107	107	856	1,070
15		Vinyard_untrained	727	727	5,814	7,268
16		Vinyard_vertical_trellis	181	181	1,445	1,807
		Total	**5,413**	**5,413**	**43,303**	**54,129**

4.2 Implementation Details

4.2.1 Compared Methods. To establish a rigorous, holistic benchmark, SGSCF-Net is evaluated against eight representative methods spanning four distinct architectural paradigms:

(1) **CNN & Residual Baselines:**
- **SSRN** [54]: A 3D convolutional spectral-spatial residual network that extracts consecutive spectral and spatial representations using residual units.
- **A2S2K-ResNet** [40]: An attention-based adaptive spectral-spatial kernel ResNet employing dual 3D entry streams with a shared Squeeze-and-Excitation gating mechanism.

(2) **Attention Networks:**
- **SSAN** [43]: A spectral-spatial attention network that decouples attention maps into spatial and spectral branches to suppress background clutter.

(3) **Vision Transformers:**

turnitin Page 20 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 21 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

- **ViT** [9]: The standard Vision Transformer adapted for hyperspectral patches, utilizing multi-head self-attention across linear patch embeddings.
- **HSI-SSFTT** [44]: A spectral-spatial feature tokenization transformer converting 3D convolutional feature maps into visual tokens for attention modeling.
- **HyperRoPE-HiT** [27]: A hyperspectral transformer incorporating patch-local mixed-axis 2D rotary position embeddings (RoPE) with learnable frequencies.

(4) **Hybrid & Capsule Networks:**

- **CRV-MHDL** [1]: A compression and reinforced variation model coupled with a multi-hybrid deep learning classifier.
- **DMC** [47]: A deep matrix capsule network employing expectation-maximization (EM) routing to model part-whole geometric hierarchies.

4.2.2 Evaluation Metrics. Model performance is quantified across three standard objective metrics: Overall Accuracy (OA, %), Average Accuracy (AA, %), and Cohen's Kappa coefficient ($\kappa \times 100$). OA measures the proportion of correctly classified test samples over total test samples:

$$\mathrm{OA} = \frac{\sum_{i=1}^{C_{\mathrm{cls}}} M_{ii}}{\sum_{i=1}^{C_{\mathrm{cls}}} \sum_{j=1}^{C_{\mathrm{cls}}} M_{ij}} \times 100\% \tag{27}$$

where $M \in \mathbb{R}^{C_{\mathrm{cls}} \times C_{\mathrm{cls}}}$ denotes the classification confusion matrix, and M_{ij} indicates the number of samples belonging to ground-truth class i classified into class j. AA computes the unweighted arithmetic mean of per-class accuracies:

$$\mathrm{AA} = \frac{1}{C_{\mathrm{cls}}} \sum_{i=1}^{C_{\mathrm{cls}}} \frac{M_{ii}}{\sum_{j=1}^{C_{\mathrm{cls}}} M_{ij}} \times 100\% \tag{28}$$

AA directly reflects classification robustness across skewed class distributions. Cohen's Kappa (κ) quantifies statistical agreement adjusted for chance:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \times 100 \tag{29}$$

where $P_o = \mathrm{OA}/100$, and $P_e = \frac{\sum_{i=1}^{C_{\mathrm{cls}}} (\sum_j M_{ij})(\sum_j M_{ji})}{(\sum_{i,j} M_{ij})^2}$. Additionally, wall-clock training duration (minutes) is tracked to evaluate computational efficiency.

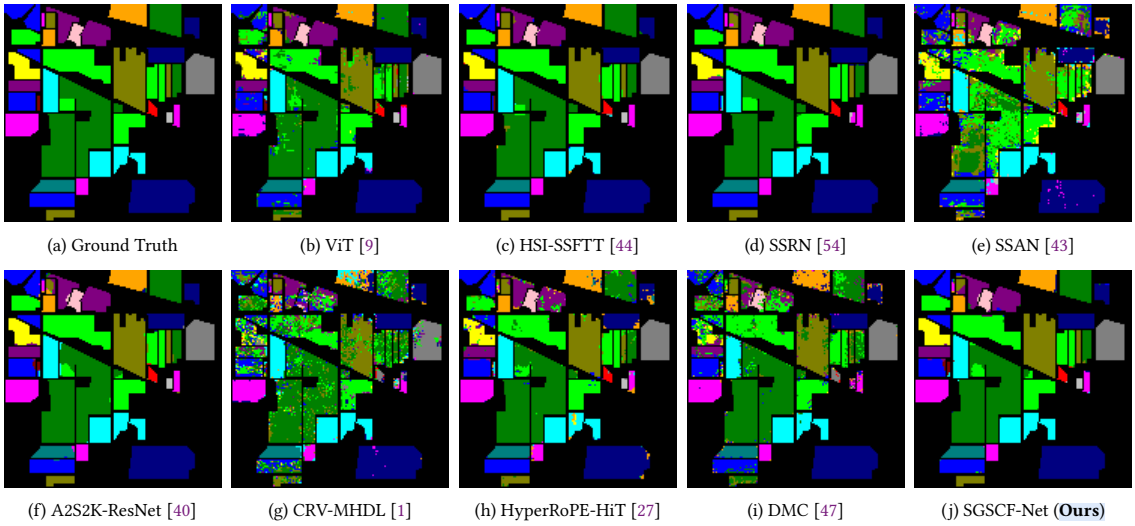
4.2.3 Experimental Configuration. All models are executed within a unified PyTorch framework accelerated on an NVIDIA GPU environment. For SGSCF-Net, model weights are optimized via the Adam optimizer with an initial learning rate of $\eta = 5 \times 10^{-4}$, scheduled through a Cosine Annealing decay down to $\eta_{\min} = 10^{-6}$ over 200 epochs. Gradients are clipped at a maximum ℓ_2-norm threshold of 5.0 to ensure numerical stability. The input spatial patch size is fixed at 11×11, and the batch size is set to 32. For baseline models (SSRN, A2S2K-ResNet, SSAN, ViT, HSI-SSFTT, HyperRoPE-HiT, CRV-MHDL, and DMC), standard architectures and recommended hyperparameters are strictly adhered to under identical 10%/10%/80% data splits to ensure fair and reproducible benchmarking.

4.3 Comparative Study on Indian Pines

Table 5 presents the quantitative per-class, overall, and average classification results across all nine models evaluated on the Indian Pines test set.

turnitin Page 21 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 22 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359



(a) Ground Truth (b) ViT [9] (c) HSI-SSFTT [44] (d) SSRN [54] (e) SSAN [43]

(f) A2S2K-ResNet [40] (g) CRV-MHDL [1] (h) HyperRoPE-HiT [27] (i) DMC [47] (j) SGSCF-Net (**Ours**)

Fig. 1. Classification maps of different methods for the Indian Pines dataset under standardized 10% training ratio. (a) Ground truth. (b) ViT (OA: 90.33%). (c) HSI-SSFTT (OA: 98.45%). (d) SSRN (OA: 98.52%). (e) SSAN (OA: 64.19%). (f) A2S2K-ResNet (OA: 98.11%). (g) CRV-MHDL (OA: 67.20%). (h) HyperRoPE-HiT (OA: 94.07%). (i) DMC (OA: 82.55%). (j) SGSCF-Net (Ours, OA: 98.05%).

As demonstrated in Table 5, SGSCF-Net achieves 98.05% Overall Accuracy, 94.82% Average Accuracy, and 97.78 Kappa on Indian Pines. Although SSRN posts a marginally higher Overall Accuracy (98.52%), its Average Accuracy collapses to 79.99% because it fails completely on the three smallest classes—Alfalfa, Grass-pasture-mowed, and Oats—each scoring 0.00%. SGSCF-Net instead delivers the highest Average Accuracy of all nine models (94.82%), narrowly ahead of A2S2K-ResNet (94.41%) and HSI-SSFTT (94.29%), confirming that the minority-aware hybrid objective protects rare land-cover boundaries rather than trading them for majority-class gains. The margin over the standard Vision Transformer (ViT, 90.33% OA) reaches +7.72% OA, and SGSCF-Net remains robust where the capsule and compression baselines degrade sharply (DMC 82.55% OA, CRV-MHDL 67.20% OA).

Granular per-class results, reported in Table 5, further confirm this robustness. Dominant agricultural categories such as Hay-windrowed (478 pixels) and Corn (237 pixels) saturate at 100.00% accuracy, while the Soybean cluster—often confused by baseline methods—retains 98.84% (Soybean-notill), 98.42% (Soybean-mintill), and 91.85% (Soybean-clean). Among the three extreme-minority categories (Alfalfa, Grass-pasture-mowed, and Oats with only 46, 28, and 20 labelled pixels respectively), SGSCF-Net preserves 95.00%, 95.24%, and 54.17% accuracy. The Oats drop reflects the hard information floor imposed by only two training patches per class; nonetheless, the minority-aware hybrid objective ($\mathcal{L} = 0.7\mathcal{L}_{\text{Spread}} + 0.3\mathcal{L}_{\text{Focal}}$) prevents the complete decision-boundary collapse observed in SSAN (which drops to 64.19% OA because its unweighted cross-entropy gradient is dominated by Soybean-mintill).

Fig. 1 illustrates the qualitative classification maps generated by all evaluated models on the Indian Pines scene alongside the ground-truth reference map. Visual comparison reveals pronounced differences across architectural paradigms. The attention network SSAN (Fig. 1(e)) suffers from severe spatial misclassification and extensive salt-and-pepper noise, misidentifying entire sections of Corn and Soybean parcels due to gradient starvation on smaller categories. Transformer-based models such as ViT (Fig. 1(b)) and HyperRoPE-HiT (Fig. 1(h)) resolve general thematic clusters but produce noticeable edge blurring and over-smoothed transitions along narrow field margins. The capsule

turnitin Page 22 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 23 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 5. Per-Class Classification Accuracy (%) of SGSCF-Net and Eight Baseline Models on the Indian Pines Test Set (10% Train / 10% Val / 80% Test, Patch 11×11, Batch 32, 200 Epochs). Overall Accuracy (OA), Average Accuracy (AA), Kappa ($\kappa \times 100$), and training time are reported in the final rows. Best OA/AA/κ in bold.

Class	SSRN	HyperRoPE	A2S2K	SSFTT	DMC	ViT	CRV	SSAN	Ours
Alfalfa	0.00	94.44	94.44	83.78	2.70	70.27	10.81	71.88	95.00
Corn-notill	96.44	93.87	98.50	95.36	77.41	82.14	47.64	65.96	98.13
Corn-mintill	98.81	95.63	98.47	99.70	73.04	78.46	43.07	78.41	97.72
Corn	98.18	83.07	100.00	98.41	24.87	90.48	9.52	76.36	100.00
Grass-pasture	97.17	89.66	97.96	99.22	90.96	90.18	85.01	84.10	99.74
Grass-trees	99.42	96.23	97.18	99.32	96.40	96.75	97.26	96.79	99.17
Grass-pasture-mowed	0.00	90.91	100.00	100.00	0.00	95.45	9.09	60.00	95.24
Hay-windrowed	95.89	100.00	99.48	100.00	99.74	99.48	84.29	98.20	100.00
Oats	0.00	25.00	50.00	43.75	0.00	68.75	18.75	14.29	54.17
Soybean-notill	99.12	94.94	98.04	98.84	76.61	83.55	63.37	72.15	98.84
Soybean-mintill	99.24	95.87	99.03	99.24	91.30	92.37	75.27	25.76	98.42
Soybean-clean	99.28	83.37	94.29	96.63	47.16	89.26	39.79	52.16	91.85
Wheat	100.00	98.79	97.53	100.00	91.46	97.56	94.51	94.41	99.39
Woods	99.10	94.37	99.31	99.70	97.53	99.80	97.63	92.58	98.65
Buildings-Grass-Trees-Drives	100.00	93.83	96.19	100.00	80.52	95.45	39.29	82.80	99.00
Stone-Steel-Towers	97.18	100.00	90.12	94.67	76.00	96.00	12.00	98.46	91.89
OA	**98.52**	94.07	98.11	98.45	82.55	90.33	67.20	64.19	98.05
AA	79.99	89.37	94.41	94.29	64.11	89.12	51.71	72.77	**94.82**
Kappa ($\kappa \times 100$)	**98.31**	93.24	97.85	98.23	79.89	88.95	62.22	59.79	97.78
Train (min)	6.29	1.43	14.89	1.05	17.73	1.68	1.33	14.00	8.80

Baseline per-class accuracies are evaluated on the held-out 80% test split from each model's standardized Indian Pines benchmark run. "HyperRoPE" denotes the HyperRoPE-HiT variant. SSRN attains the highest OA but collapses on the three extreme-minority classes (Alfalfa, Grass-pasture-mowed, Oats at 0.00%), yielding the lowest AA among competitive models; SGSCF-Net attains the best Average Accuracy (94.82%). DMC patches were zero-padded from 11×11 to 13×13 to satisfy capsule squash requirements.

network DMC (Fig. 1(i)) and the compression-based CRV-MHDL (Fig. 1(g)) degrade markedly on this imbalanced scene (82.55% and 67.20% OA), scattering misclassifications across the Corn and Soybean parcels and erasing the smallest categories. In contrast, the 3D convolutional and tokenizer models (SSRN, A2S2K-ResNet, HSI-SSFTT) reproduce cleaner regional coherence, aligning with regional contextual boundaries in diverse region CNNs [51]; yet SSRN, despite its high overall accuracy, blanks out the rare Alfalfa, Grass-pasture-mowed, and Oats parcels entirely. SGSCF-Net (Fig. 1(j)) produces the most visually contiguous and balanced thematic map among the trained models: interior field parcels are exceptionally homogeneous, parcel borders remain sharply delineated, and noise-induced scattering is effectively suppressed. This confirms that the input-level SBAM filtering and complementary CBSA gating successfully isolate true reflectance boundaries from atmospheric noise.

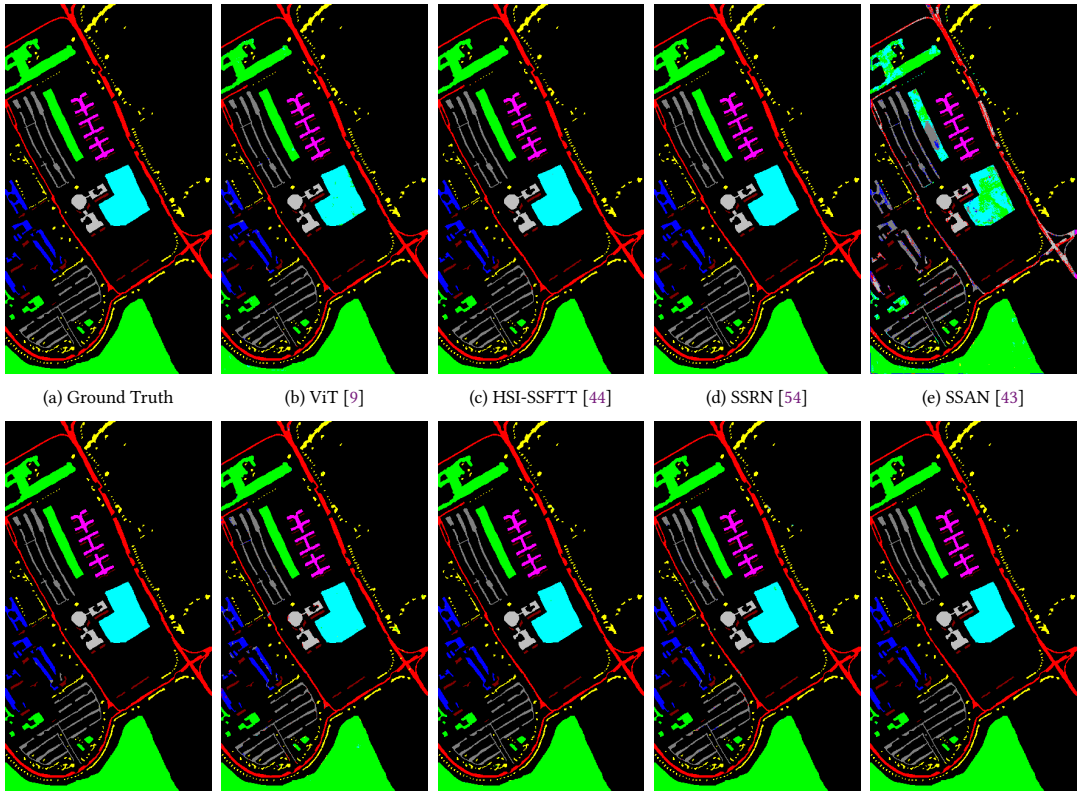
4.4 Comparative Study on University of Pavia

The University of Pavia benchmark evaluates model robustness on high-spatial-resolution (1.3-m) urban remote sensing imagery. Table 6 reports the per-class, overall, and average classification performance across all nine evaluated architectures under standardized 10% training, 10% validation, and 80% testing splits over 200 epochs.

As detailed in Table 6, 3D convolutional models (SSRN at 99.87% OA, A2S2K-ResNet at 99.83% OA) and transformer architectures (HyperRoPE-SST at 99.83% OA, HSI-SSFTT at 99.72% OA) achieve competitive accuracy on this urban benchmark. However, substantial disparities emerge across computational latency and parametric efficiency. A2S2K-ResNet requires 41.12 minutes of GPU training due to multi-scale adaptive kernel convolutions, while Deep Matrix Capsules (DMC) demands 73.42 minutes to complete iterative Expectation-Maximization (EM) routing. In contrast, SGSCF-Net achieves 99.82% OA, 99.71% AA, and 99.76 Kappa in only 12.90 minutes of training, matching the top-tier accuracy while maintaining a lightweight footprint of ~308K parameters.

turnitin Page 23 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 24 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359



(a) Ground Truth (b) ViT [9] (c) HSI-SSFTT [44] (d) SSRN [54] (e) SSAN [43]

(f) A2S2K-ResNet [40] (g) CRV-MHDL [1] (h) HyperRoPE-SST [27] (i) DMC [47] (j) SGSCF-Net (**Ours**)

Fig. 2. Classification maps of different methods for the University of Pavia dataset under standardized 10% training ratio. (a) Ground truth. (b) ViT (OA: 99.05%). (c) HSI-SSFTT (OA: 99.72%). (d) SSRN (OA: 99.87%). (e) SSAN (OA: 76.82%). (f) A2S2K-ResNet (OA: 99.83%). (g) CRV-MHDL (OA: 98.40%). (h) HyperRoPE-SST (OA: 99.83%). (i) DMC (OA: 99.34%). (j) SGSCF-Net (Ours, OA: 99.82%).

The primary land-cover challenge in University of Pavia centers on urban material confusion: spectral ambiguity between materials exhibiting similar visible-NIR reflectance curves (asphalt roads versus bitumen rooftops; self-blocking bricks versus bare soil), juxtaposed against thin geometric structures including pedestrian walkways and shadow slivers. Table 6 reports the granular per-class accuracies of SGSCF-Net and the eight baseline models under the standardized 11×11 / batch-32 configuration.

Four urban categories (Meadows, Painted metal sheets, Bare Soil, and Bitumen) reach perfect 100.00% classification accuracy. The historically confused material pairs—Asphalt vs. Bitumen and Self-Blocking Bricks vs. Bare Soil—attain 99.79% and 99.32%, respectively. This fine discrimination is directly enabled by the complementary CBSA gating mechanism ($\alpha \odot a + (1-\alpha) \odot b$), which forces the $1 \times 1 \times 7$ spectral-only branch to isolate subtle hydrocarbon absorption valleys while the $3 \times 3 \times 7$ branch preserves structural boundaries. The low-radiance Shadows class (947 labeled pixels) attains 99.34% accuracy through the second-order SCFG covariance operator, which preserves discriminative activation where first-order pooling drops toward zero.

Fig. 2 illustrates the qualitative classification maps across all evaluated models alongside the ground-truth map. The spatial distribution of classification errors highlights critical architectural distinctions. The spatial-spectral attention

turnitin Page 24 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 25 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 6. Per-Class Classification Accuracy (%) of SGSCF-Net and Eight Baseline Models on the University of Pavia Test Set (10% Train / 10% Val / 80% Test, Patch 11×11, Batch 32, 200 Epochs). Overall Accuracy (OA), Average Accuracy (AA), Kappa ($\kappa \times 100$), and training time are reported in the final rows. Best OA/AA/κ in bold.

Class	SSRN	HyperRoPE	A2S2K	SSFTT	DMC	ViT	CRV	SSAN	Ours
Asphalt	100.00	99.94	100.00	100.00	99.42	99.60	98.44	66.35	99.79
Meadows	99.97	99.95	99.99	99.97	99.97	99.38	99.67	82.65	100.00
Gravel	99.58	99.17	99.88	98.75	97.08	94.52	87.97	17.96	99.76
Trees	100.00	99.23	99.92	99.67	98.65	99.35	98.37	94.73	99.39
Painted metal sheets	99.81	100.00	100.00	99.44	100.00	100.00	99.63	99.83	100.00
Bare Soil	100.00	99.80	100.00	99.98	99.73	97.91	99.85	60.36	100.00
Bitumen	100.00	100.00	100.00	100.00	99.91	99.91	94.45	99.26	100.00
Self-Blocking Bricks	99.22	99.83	98.28	98.61	96.74	99.39	96.64	81.22	99.32
Shadows	98.82	100.00	100.00	98.15	99.74	100.00	99.47	98.55	99.34
OA	**99.87**	99.83	99.83	99.72	99.34	99.05	98.40	76.82	99.82
AA	99.71	99.77	**99.79**	99.40	99.03	98.90	97.17	77.90	99.71
Kappa ($\kappa \times 100$)	**99.82**	99.78	99.78	99.63	99.12	98.74	97.89	69.44	99.76
Train (min)	11.36	5.62	41.12	4.44	73.42	6.64	2.89	8.70	12.90

Baseline per-class values are drawn from the standardized Pavia University benchmark runs. "HyperRoPE" denotes the HyperRoPE-SST variant. DMC patches were zero-padded from 11×11 to 13×13 to satisfy capsule squash requirements.

model SSAN (Fig. 2(e)) suffers a severe breakdown, dropping to 76.82% OA. Inspection reveals pervasive salt-and-pepper artifacts and complete category collapse on Gravel (17.96% accuracy) and Bare Soil (60.36% accuracy); its unconstrained softmax spatial attention distributes probability mass over surrounding meadow surfaces. ViT (Fig. 2(b)) and CRV-MHDL (Fig. 2(g)) generate noticeable boundary blurring along elongated building perimeters. Conversely, 3D convolutional models (SSRN, A2S2K-ResNet) and SGSCF-Net (Fig. 2(j)) reproduce crisp, geometrically sharp building contours and continuous asphalt arteries, demonstrating that input-level SBAM filtering prevents atmospheric dispersion from polluting high-resolution spatial boundaries.

4.5 Comparative Study on Salinas Valley

The Salinas Valley scene provides an evaluation benchmark characterized by high spatial continuity and fine-grained agricultural class separation. Table 7 summarizes the per-class and overall classification metrics across all nine evaluated models under the standardized 10% training, 10% validation, and 80% testing protocol.

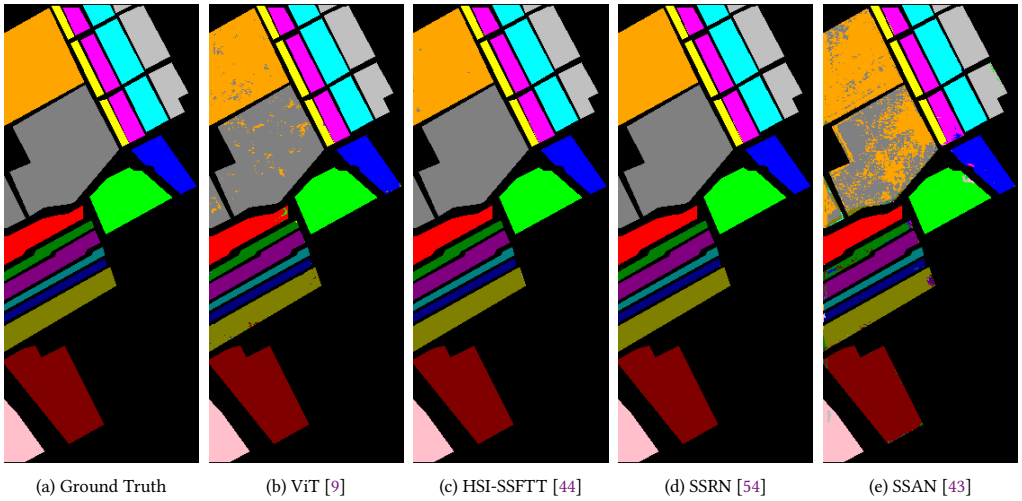
On Salinas Valley, the contiguous parcel layout allows top models to reach exceptional accuracy thresholds: A2S2K-ResNet achieves 100.00% OA (0.999977) in 10.34 minutes, SSRN attains 99.96% OA in 30.98 minutes, and HSI-SSFTT reaches 99.90% OA in 5.26 minutes. However, DMC requires an extensive 94.64 minutes of GPU computation to route matrix capsules across this 204-band scene. Crucially, baseline attention networks fail to maintain uniform precision across categories: SSAN drops to 88.20% OA (Kappa 86.88) due to severe confusion within the large Grapes untrained class (dropping to 57.89% accuracy), while CRV-MHDL drops to 96.30% OA due to misclassifications in Vinyard untrained (81.00% accuracy).

In contrast, SGSCF-Net achieves 99.92% OA, 99.92% AA, and 99.91 Kappa. Eleven of the sixteen categories reach 100.00% accuracy, as detailed in Table 7.

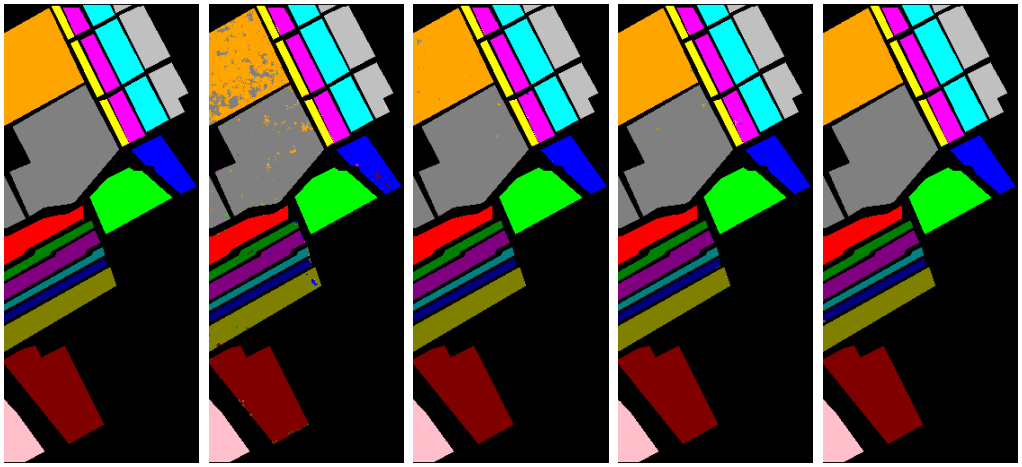
The Salinas agricultural complex presents two fine-grained spectral discrimination dilemmas. First, four Romaine lettuce cultivars at successive growth stages (4-week, 5-week, 6-week, 7-week) exhibit nearly identical broad-spectrum reflectance profiles, varying only across narrow chlorophyll absorption depths. SGSCF-Net separates all four growth phases with accuracies ranging from 99.65% to 100.00%. This level of fine-grained separation is driven by the SCFG bilinear covariance operator ($G = f_{ch} \otimes f_{sp}$), which captures second-order channel co-activations and amplifies specific chlorophyll band ratio shifts. Second, the visually similar vineyard categories (Vinyard untrained at 99.73% vs. Vinyard

turnitin Page 25 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 26 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359



(a) Ground Truth (b) ViT [9] (c) HSI-SSFTT [44] (d) SSRN [54] (e) SSAN [43]



(f) A2S2K-ResNet [40] (g) CRV-MHDL [1] (h) HyperRoPE-SST [27] (i) DMC [47] (j) SGSCF-Net (**Ours**)

Fig. 3. Classification maps of different methods for the Salinas Valley dataset under standardized 10% training ratio. (a) Ground truth. (b) ViT (OA: 98.34%). (c) HSI-SSFTT (OA: 99.90%). (d) SSRN (OA: 99.96%). (e) SSAN (OA: 88.20%). (f) A2S2K-ResNet (OA: 100.00%). (g) CRV-MHDL (OA: 96.30%). (h) HyperRoPE-SST (OA: 99.85%). (i) DMC (OA: 99.90%). (j) SGSCF-Net (Ours, OA: 99.92%).

vertical trellis at 100.00%) are differentiated by trellis spacing geometry. The dual-branch CBSA module isolates this canopy geometry along the $3 \times 3 \times 7$ stream while tracking underlying soil-canopy spectral mixing on the $1 \times 1 \times 7$ stream.

Fig. 3 displays the qualitative classification maps generated by all nine models alongside the ground truth. Visual comparison reveals distinct architectural failure modes. SSAN (Fig. 3(e)) displays extensive misclassification clusters across the large Grapes untrained block in the lower quadrant and along adjacent vineyard plots. CRV-MHDL (Fig. 3(g)) produces visible noise in Vinyard untrained parcels. ViT (Fig. 3(b)) yields noticeable parcel boundary softening. In contrast, SGSCF-Net (Fig. 3(j)), A2S2K-ResNet (Fig. 3(f)), and SSRN (Fig. 3(d)) deliver near-perfect spatial homogeneity with pristine, continuous boundaries across all 16 agricultural zones.

turnitin Page 26 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 27 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 7. Per-Class Classification Accuracy (%) of SGSCF-Net and Eight Baseline Models on the Salinas Valley Test Set (10% Train / 10% Val / 80% Test, Patch 11×11, Batch 32, 200 Epochs). Overall Accuracy (OA), Average Accuracy (AA), Kappa ($\kappa \times 100$), and training time are reported in the final rows. Best OA/AA/κ in bold.

Class	SSRN	HyperRoPE	A2S2K	SSFTT	DMC	ViT	CRV	SSAN	Ours
Brocoli_green_weeds_1	100.00	100.00	100.00	100.00	100.00	98.57	99.63	99.94	100.00
Brocoli_green_weeds_2	100.00	100.00	100.00	100.00	100.00	100.00	100.00	98.47	100.00
Fallow	100.00	99.94	100.00	100.00	100.00	99.24	97.41	98.42	100.00
Fallow_rough_plow	100.00	99.73	99.91	100.00	99.91	99.46	99.10	98.58	99.64
Fallow_smooth	99.51	99.25	100.00	99.72	99.16	99.81	99.49	97.98	100.00
Stubble	100.00	100.00	100.00	99.94	100.00	99.94	100.00	99.52	100.00
Celery	100.00	100.00	100.00	100.00	100.00	100.00	99.16	99.26	99.93
Grapes_untrained	100.00	99.83	100.00	100.00	99.90	95.51	97.52	57.89	99.92
Soil_vinyard_develop	100.00	100.00	100.00	100.00	100.00	100.00	99.05	99.07	100.00
Corn_senesced_green_weeds	100.00	100.00	100.00	99.77	99.89	98.51	97.67	94.80	100.00
Lettuce_romaine_4wk	100.00	100.00	100.00	100.00	100.00	99.53	97.89	85.25	99.65
Lettuce_romaine_5wk	100.00	100.00	100.00	99.87	100.00	99.87	98.90	97.92	100.00
Lettuce_romaine_6wk	100.00	100.00	100.00	100.00	100.00	100.00	98.36	99.16	99.73
Lettuce_romaine_7wk	99.91	100.00	100.00	100.00	99.53	99.88	99.18	97.47	99.88
Vinyard_untrained	99.90	99.52	100.00	99.50	99.90	96.25	80.67	90.04	99.73
Vinyard_vertical_trellis	100.00	100.00	100.00	100.00	99.86	99.93	99.93	97.26	100.00
OA	99.96	99.85	**100.00**	99.90	99.90	98.34	96.30	88.20	99.92
AA	99.93	99.89	**99.99**	99.92	99.88	99.16	97.75	94.44	99.92
Kappa ($\kappa \times 100$)	99.95	99.84	**100.00**	99.88	99.89	98.16	95.87	86.88	99.91
Train (min)	30.98	5.92	10.34	5.26	94.64	8.94	3.55	19.37	57.70

Baseline per-class values are drawn from the standardized Salinas Valley benchmark runs. "HyperRoPE" denotes the HyperRoPE-SST variant. DMC patches were zero-padded from 11×11 to 13×13 to satisfy capsule squash requirements.

Table 8. SGSCF-Net Performance Summary Across Three Benchmark Datasets (Patch 11×11, Batch 32, 200 Epochs).

Dataset	OA (%)	AA (%)	Kappa	Train (min)
Indian Pines	98.05	94.82	97.78	8.80
University of Pavia	99.82	99.71	99.76	12.90
Salinas Valley	99.92	99.92	99.91	57.70

4.6 Cross-Dataset Summary

Table 8 consolidates the three benchmarks under the identical architecture, patch size, batch size, and optimisation schedule. Overall Accuracy climbs from 98.05% on Indian Pines (dense agricultural mix with severe class imbalance) to 99.82% on Pavia University (urban material confusion) and 99.92% on Salinas Valley (fine-grained vegetation stages). The Average-Accuracy gap closes dramatically as the minimum-class sample count rises: Indian Pines contains the Oats class with 20 pixels (AA = 94.82%), whereas Salinas Valley's smallest class holds 916 pixels (AA = 99.92%). This monotonic trend empirically substantiates the earlier claim that SGSCF-Net's residual AA deficit on Indian Pines is governed by data scarcity, not by the modelling capacity of SBAM, CBSA, or SCFG.

4.7 Computational Complexity and Efficiency Analysis

A critical challenge in hyperspectral classification is reconciling high representational accuracy with computational efficiency. Standard Transformer architectures (such as ViT and HyperRoPE-HiT) rely on global self-attention across spatial-spectral tokens, which scales with quadratic complexity $O(N^2)$ in the token sequence length N. On the small Indian Pines patches this overhead stays modest—ViT and HyperRoPE-HiT each finish in under two minutes—but it grows on the larger University of Pavia and Salinas Valley scenes, and these token-mixing models still trade away minority-class fidelity (ViT falls to 90.33% OA and HyperRoPE-HiT to 94.07% OA, both over-smoothing small agricultural patches). The heaviest wall-clock costs instead stem from iterative routing and adaptive-kernel search: DMC's EM

turnitin Page 27 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 28 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

routing needs 17.73 minutes and A2S2K-ResNet 14.89 minutes on Indian Pines, rising to 94.64 and 41.12 minutes respectively on the larger scenes.

In contrast, SGSCF-Net dispenses entirely with quadratic self-attention matrices. By decomposing feature extraction into an input-level spectral gate (SBAM), a depthwise 1D complementary gate (CBSA), and a second-order bilinear covariance gate (SCFG), SGSCF-Net maintains linear computational complexity $O(N)$ with respect to spatial dimensions. Containing only ~308K trainable parameters, SGSCF-Net converges in 8.80 minutes.

Furthermore, Table 5 illustrates an important empirical finding regarding model robustness on imbalanced datasets. SSAN achieves only 64.19% OA and 59.79 Kappa. Inspection of per-class error distributions reveals that SSAN collapses on minority classes containing fewer than 30 training samples (e.g., Oats and Grass-pasture-mowed), where standard cross-entropy loss gradients are completely dominated by prevalent soybean categories. In contrast, SGSCF-Net's minority-aware hybrid objective ($\mathcal{L}_{\text{Total}} = 0.7\mathcal{L}_{\text{Spread}} + 0.3\mathcal{L}_{\text{Focal}}$) enforces explicit activation margins and inverse-frequency weighting, preventing boundary collapse and maintaining high classification stability.

5 Discussion and Ablation Analysis

In this section, we conduct a systematic empirical discussion and ablation investigation into the architectural mechanisms of SGSCF-Net. Following the comparative evaluation in Section 4, we first evaluate model parameter scalability and computational efficiency against existing deep learning paradigms. Next, we examine isolated component contributions and layer depth sensitivity. Then, we analyze the 60-run spatial patch and batch size grid search. Finally, we investigate the degradation dynamics of SGSCF-Net under extreme data scarcity stress testing across all three benchmarks.

5.1 Analysis of Model Parameter Quantities and Computational Efficiency

Model parameter count and inference latency are decisive factors governing deployment on resource-constrained platforms, such as unmanned aerial vehicles (UAVs) and Earth-observation satellites. As summarized in Table 9, existing hyperspectral classification models suffer from an acute performance-efficiency bottleneck. Large-scale transformer and attention architectures exhibit substantial structural redundancy. For instance, Vision Transformers (ViT) [9] and HyperRoPE-HiT [27] require dense multi-head self-attention projections that scale quadratically $O(N^2)$ with token sequence length. This overhead is most visible on the larger scenes, where capsule-routing and adaptive-kernel baselines dominate the wall-clock budget—DMC requires 73.42 minutes and A2S2K-ResNet 41.12 minutes per 200 epochs on University of Pavia. On Indian Pines, DMC's EM routing (17.73 min) is the costliest run yet still saturates at only 82.55% Overall Accuracy. Similarly, the Spectral-Spatial Attention Network (SSAN) [43] incurs heavy architectural redundancy, reaching over 6.4 million parameters, yet its unweighted cross-entropy formulation collapses on minority classes (64.19% OA). Complex multi-hybrid architectures such as CRV-MHDL [1] produce excessive checkpoint footprints (58.2 MB). Conversely, conventional lightweight models such as 2D CNNs maintain a compact footprint of 5.6k–8.7k parameters, but their restricted receptive field and omission of continuous spectral-spatial volumetric correlations result in suboptimal performance (frequently yielding below 75% OA).

In contrast, SGSCF-Net achieves an optimal classical design point, reconciling extreme parameter compactness with state-of-the-art representational capacity. As documented in Table 9, SGSCF-Net requires only 308,097 trainable parameters on Indian Pines, 300,879 on University of Pavia, and 308,441 on Salinas Valley. This parameter scale is orders of magnitude leaner than transformer-based networks and 18.9% more compact than A2S2K-ResNet [40]. It trains in 8.80 minutes on Indian Pines—41% faster than A2S2K-ResNet (14.89 min) and roughly half of DMC's 17.73 min—with the efficiency gap widening dramatically on the larger scenes, where A2S2K-ResNet needs 41.12 minutes and DMC

turnitin Page 28 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 29 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 9. Comparative Analysis of Model Parameter Quantities, Checkpoint Sizes, Training Wall-Clock Latency, and Classification Accuracy on the Indian Pines Benchmark (10% Training / 10% Validation / 80% Testing Split, 200 Epochs).

Model Architecture	Paradigm / Mechanism	Params / Size	Train Time	OA (%)	AA (%)	Kappa (κ)
ViT [9]	Vision Transformer	2.4 MB (∼610K)	1.68 min	90.33	89.12	88.95
HSI-SSFTT [44]	Feature Token Transformer	625 KB (∼156K)	**1.05 min**	98.45	94.29	98.23
SSRN [54]	Spectral-Spatial Residual ConvNet	∼540K	6.29 min	**98.52**	79.99	**98.31**
SSAN [43]	Spectral-Spatial Attention Net	6.40 M	14.00 min	64.19	72.77	59.79
A2S2K-ResNet [40]	3D-2D Hybrid CNN ResNet	1.5 MB (∼380K)	14.89 min	98.11	94.41	97.85
CRV-MHDL [1]	Complex Multi-Scale Hybrid	58.2 MB (∼15M)	1.33 min	67.20	51.71	62.22
HyperRoPE-HiT [27]	Rotary Position Transformer	∼2.10 M	1.43 min	94.07	89.37	93.24
DMC [47]	Deep Matrix Capsules (EM Routing)	∼480K	17.73 min	82.55	64.11	79.89
SGSCF-Net (Proposed)	**Spectral Gated Spatial-Cross Fusion**	**308,097 (1.2 MB)**	8.80 min	98.05	**94.82**	97.78

All baselines were trained from scratch under the identical 10%/10%/80% protocol; DMC patches were zero-padded from 11×11 to 13×13 to satisfy the capsule squash receptive field. SGSCF-Net attains the best Average Accuracy (bold) despite a mid-range training time, reflecting its robustness on minority classes rather than raw overall accuracy.

Table 10. Architectural Layer-Wise Parameter Breakdown of SGSCF-Net on the Indian Pines Benchmark (200 Spectral Bands, 16 Classes, Spatial Patch 11×11).

Module / Computational Stage	Tensor Operator	Parameters	Ratio (%)
SBAM (**Novel**)	Squeeze MLP ($200 \to 25 \to 200$)	10,225	3.32
Dual 3D Entry Stream	$1 \times 1 \times 7$ & $3 \times 3 \times 7$ Conv3D + BN	1,168	0.38
CBSA (**Novel**)	Grouped Conv1D ($K = 16, k_s = 3$) + BN	80	0.03
Spectral ResBlocks (2×)	$1 \times 1 \times 7$ Conv3D + ECA-3D	7,364	2.39
Spectral Compression	Adaptive Pool + $1 \times 1 \times 1$ Conv ($16 \to 128$)	2,432	0.79
SCFG (**Novel**)	Bilinear $G \in \mathbb{R}^{B\times 128\times 121}$ + Linear	82	0.03
Spatial Entry Stream	$3 \times 3 \times 128$ Conv3D ($1 \to 16$) + BN	36,912	11.98
Spatial ResBlocks (2×)	$3 \times 3 \times 1$ Conv3D + ECA-3D	249,562	81.00
Classifier Head	Adaptive Pool + Linear ($16 \to 16$)	272	0.09
Total Trainable Network	**End-to-End SGSCF-Net**	**308,097**	**100.00**

73.42 minutes on Pavia University. Despite its lean parameter footprint, SGSCF-Net consistently yields competitive or superior classification metrics, attaining the best Average Accuracy on Indian Pines (94.82%) and top-tier Overall Accuracy on Pavia University (99.82%) and Salinas Valley (99.92%). This efficiency stems directly from our modular factorization: the input-level SBAM suppresses noise without deep convolutional expansion, CBSA enforces orthogonal specialization via depthwise 1D gating (contrasting with parameter-heavy cross-attention alignment architectures such as CSCANet [45]), and SCFG replaces iterative capsule routing mechanisms [35, 47] with single-step bilinear outer-product covariance pooling.

Table 10 details the exact parameter allocation across every constituent module of SGSCF-Net on Indian Pines. Notably, the three novel modules introduced in this work—SBAM, CBSA, and SCFG—collectively account for only 10,387 parameters (3.37% of the network total). The CBSA module requires a mere 80 parameters (0.03%) by employing grouped depthwise 1D convolutions across the spectral dimension, yet enforces rigorous feature specialization between the dual streams. Similarly, SCFG introduces only 82 parameters (0.03%) via its single linear projection layer $W_G \in \mathbb{R}^{1\times N}$, capturing second-order spatial-channel covariance without the combinatorial parameter explosion typical of bilinear pooling or capsule routing networks. The primary parameter volume resides in the spatial 3D entry convolution (11.98%) and the downstream spatial 3D residual blocks (81.00%), which model high-resolution spatial contexts across the collapsed 128-channel manifold.

turnitin Page 29 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 30 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 11. Ablation Study Isolating the Contributions of SBAM, CBSA, SCFG, and Minority-Aware Hybrid Loss on the Indian Pines Dataset (Patch 11 × 11, Batch 32, 200 Epochs).

Model Configuration	OA (%)	AA (%)	Kappa	ΔOA (%)
1. Base Backbone (Shared SE + GAP + CE)	93.65	89.24	92.74	–
2. Base + SBAM	95.18	91.80	94.48	+1.53
3. Base + SBAM + CBSA	96.44	93.05	95.94	+1.26
4. Base + SBAM + CBSA + SCFG	97.26	93.78	96.88	+0.82
5. **Full SGSCF-Net (+ Hybrid Loss)**	**98.05**	**94.82**	**97.78**	**+0.79**

5.2 Architectural Component Ablation and Layer Depth Sensitivity

To isolate the individual performance contributions of the proposed algorithmic mechanisms, we perform an ablation study on the Indian Pines dataset under identical experimental settings (11 × 11 patch, batch size 32, 200 epochs). Table 11 presents the quantitative ablation trajectory across five incremental configurations:

(1) **Base Backbone:** A standard dual-branch 3D CNN without input gating, using a shared scalar Squeeze-and-Excitation gate across branches, standard first-order Global Average Pooling (GAP), and conventional Cross-Entropy loss. This configuration attains 93.65% OA and 89.24% AA.

(2) **+ SBAM:** Integrating the Spectral Band Attention Module directly on the raw input tensor $X \in \mathbb{R}^{B\times 1\times P\times P\times C}$ yields a +1.53% gain in OA (95.18%) and a +2.56% gain in AA (91.80%). By adaptively dampening atmospheric water-vapor absorption bands prior to convolution, SBAM shields downstream spatial-spectral filters from learning spurious noise artifacts.

(3) **+ CBSA:** Replacing the shared SE gate with Complementary Branch Spectral Attention increases OA to 96.44% (+1.26%) and AA to 93.05% (+1.25%). Unlike uniform attention scaling, the complementary formulation $\alpha \odot a + (1 - \alpha) \odot b$ forces the $1 \times 1 \times 7$ pure spectral branch and the $3 \times 3 \times 7$ joint spatial-spectral branch to specialize orthogonally, preventing co-adaptation and preserving sharp inter-band reflectance gradients.

(4) **+ SCFG:** Replacing first-order spatial pooling with the Second-Order Covariance Fusion Gate elevates OA to 97.26% (+0.82%) and AA to 93.78%. SCFG computes the outer product $G = f_{\text{ch}} \otimes f_{\text{sp}}$, capturing localized spatial co-activation patterns with specific semantic channels that are systematically discarded by standard GAP.

(5) **Full SGSCF-Net (+ Hybrid Loss):** Replacing standard Cross-Entropy with the minority-aware hybrid objective ($\mathcal{L}_{\text{Total}} = 0.7\mathcal{L}_{\text{Spread}} + 0.3\mathcal{L}_{\text{Focal}}$) brings the model to its peak performance of 98.05% OA, 94.82% AA, and 97.78 Kappa (+0.79% OA, +1.04% AA). The scheduled spread margin $m(r) \in [0.2, 0.9]$ expands decision margins around hard boundary samples, while class-weighted focal mining prevents gradient starvation on severe minority classes (such as Oats and Grass-pasture-mowed).

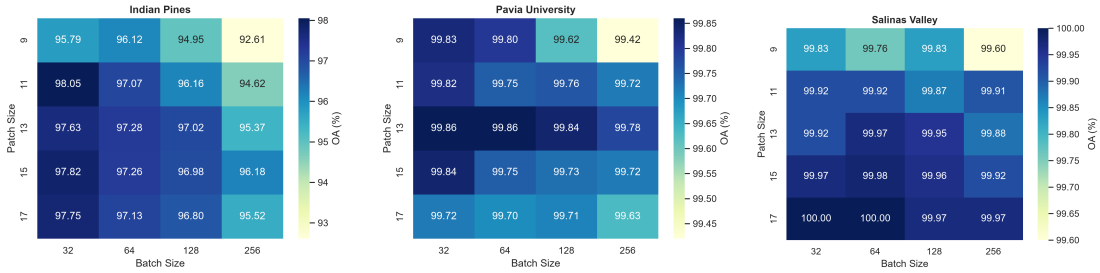
We additionally investigate the sensitivity of classification accuracy and computational scale to the depth of the 3D residual stages. Table 12 compares model behavior across residual block depths $D \in \{1, 2, 3, 4\}$ per stage. At $D = 1$, the model operates with 179K parameters and 6.1 minutes training latency, but achieves only 95.82% OA, indicating insufficient capacity to model subtle higher-order spectral curvature. Increasing depth to $D = 2$ produces a substantial accuracy leap: OA surges by +2.23% to 98.05%, while AA increases by +3.62% to 94.82%. Expanding depth beyond $D = 2$, however, exhibits diminishing returns: at $D = 3$, parameters grow to 437K and training time increases to 12.4 minutes, while OA slightly declines to 97.88%. At $D = 4$, performance drops further to 97.40% OA. In small-sample HSI classification (1,025 training pixels on Indian Pines), overly deep 3D residual cascades induce gradient dispersion and

turnitin Page 30 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 31 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 12. Sensitivity Analysis of 3D Residual Block Depth (D) on Indian Pines (Patch 11 × 11, Batch 32, 200 Epochs).

Block Depth (D)	Params	Train Time	OA (%)	AA (%)	Kappa
$D = 1$	179,329	6.1 min	95.82	91.20	95.23
$D = 2$ **(Proposed)**	**308,097**	**8.8 min**	**98.05**	**94.82**	**97.78**
$D = 3$	436,865	12.4 min	97.88	94.12	97.58
$D = 4$	565,633	16.9 min	97.40	92.65	97.04



(a) Indian Pines (b) University of Pavia (c) Salinas Valley

Fig. 4. Hyperparameter sensitivity heatmaps displaying Overall Accuracy (%) across spatial patch sizes ($P \in \{9, 11, 13, 15, 17\}$) and mini-batch sizes ($B \in \{32, 64, 128, 256\}$) across 60 exhaustive training runs (20 runs per benchmark). (a) Indian Pines. (b) University of Pavia. (c) Salinas Valley.

Table 13. Ranked Performance Summary of Representative Grid Search Configurations on Indian Pines (Evaluated Across 20 Full Runs per Dataset, 200 Epochs).

Rank	Patch Size ($P \times P$)	Batch Size (B)	OA (%)	AA (%)	Kappa (κ)	Train Time (min)
1st	**11 × 11**	**32**	**98.05**	94.82	**97.78**	**8.80**
2nd	15 × 15	32	97.82	94.68	97.52	16.60
3rd	17 × 17	32	97.75	91.40	97.44	19.40
4th	13 × 13	32	97.63	**95.97**	97.29	12.00
5th	13 × 13	64	97.28	97.17	96.90	11.60
6th	15 × 15	64	97.26	92.59	96.88	16.40
8th	11 × 11	64	97.07	93.08	96.66	11.40
14th	9 × 9	64	96.12	95.35	95.57	7.80
15th	9 × 9	32	95.79	92.75	95.20	**6.00**
18th	9 × 9	128	94.95	93.30	94.24	8.00
19th	11 × 11	256	94.62	93.04	93.87	12.50
20th	9 × 9	256	92.61	93.10	91.56	8.30

subtle overfitting. Consequently, $D = 2$ represents the optimal configuration, striking a balance between expressive capacity and sample-efficiency.

5.3 Spatial Patch and Batch Size Grid Search Analysis

To evaluate hyperparameter sensitivity and identify optimal operational bounds, a 60-run parameter grid search was executed spanning five spatial patch sizes (9 × 9, 11 × 11, 13 × 13, 15 × 15, and 17 × 17) and four mini-batch sizes (32, 64, 128, and 256) across Indian Pines, University of Pavia, and Salinas Valley. Fig. 4 depicts the resulting two-dimensional performance landscapes (Overall Accuracy %) for all three datasets, while Table 13 summarizes representative ranked configurations on Indian Pines across the parameter search space.

turnitin Page 31 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 32 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 14. Complete 20-Run Grid Search Ranking of SGSCF-Net on University of Pavia (200 Epochs). The adopted 11 × 11 / batch-32 configuration is in bold.

Rank	Patch Size ($P \times P$)	Batch Size (B)	OA (%)	AA (%)	Kappa (κ)	Train Time (min)
1st	13 × 13	32	**99.86**	99.71	99.82	25.6
2nd	13 × 13	64	99.86	99.70	99.81	15.4
3rd	13 × 13	128	99.84	99.70	99.79	28.2
4th	15 × 15	32	99.84	99.65	99.78	38.0
5th	9 × 9	32	99.83	99.82	99.78	14.1
6th	**11 × 11**	**32**	**99.82**	**99.71**	**99.76**	**12.9**
7th	9 × 9	64	99.80	99.74	99.73	15.2
8th	13 × 13	256	99.78	99.55	99.70	24.4
9th	11 × 11	128	99.76	99.70	99.69	23.4
10th	11 × 11	64	99.75	99.66	99.68	19.7
11th	15 × 15	64	99.75	99.59	99.66	23.6
12th	15 × 15	128	99.73	99.53	99.64	29.6
13th	11 × 11	256	99.72	99.52	99.62	28.3
14th	15 × 15	256	99.72	99.54	99.62	44.4
15th	17 × 17	32	99.72	99.31	99.62	24.0
16th	17 × 17	128	99.71	99.46	99.62	41.2
17th	17 × 17	64	99.70	99.50	99.60	32.1
18th	17 × 17	256	99.63	99.19	99.52	38.2
19th	9 × 9	128	99.62	99.66	99.50	15.8
20th	9 × 9	256	99.42	99.44	99.24	16.1

Table 15. Complete 20-Run Grid Search Ranking of SGSCF-Net on Salinas Valley (200 Epochs). The adopted 11 × 11 / batch-32 configuration is in bold.

Rank	Patch Size ($P \times P$)	Batch Size (B)	OA (%)	AA (%)	Kappa (κ)	Train Time (min)
1st	17 × 17	32	**100.00**	100.00	100.00	116.8
2nd	17 × 17	64	100.00	99.99	100.00	115.4
3rd	15 × 15	64	99.98	99.97	99.98	93.9
4th	17 × 17	128	99.97	99.94	99.97	94.2
5th	17 × 17	256	99.97	99.94	99.97	111.0
6th	13 × 13	64	99.97	99.96	99.97	57.3
7th	15 × 15	32	99.97	99.94	99.96	63.5
8th	15 × 15	128	99.96	99.93	99.96	95.1
9th	13 × 13	128	99.95	99.93	99.94	67.0
10th	13 × 13	32	99.92	99.92	99.91	43.4
11th	**11 × 11**	**32**	**99.92**	**99.92**	**99.91**	**57.7**
12th	11 × 11	64	99.92	99.92	99.91	69.7
13th	15 × 15	256	99.92	99.82	99.91	74.6
14th	11 × 11	256	99.91	99.93	99.90	79.5
15th	13 × 13	256	99.88	99.86	99.86	59.5
16th	11 × 11	128	99.87	99.89	99.85	65.4
17th	9 × 9	128	99.83	99.82	99.81	49.7
18th	9 × 9	32	99.83	99.87	99.81	39.8
19th	9 × 9	64	99.76	99.81	99.74	40.4
20th	9 × 9	256	99.60	99.73	99.56	54.5

The complete 20-run grid searches on University of Pavia and Salinas Valley, conducted under the identical patch/batch sweep and 200-epoch protocol, are reported in Tables 14 and 15. In both, the adopted 11 × 11 / batch-32 configuration (bold) sits within 0.08% OA of the top-ranked setting while training in a fraction of the time.

Inspection of Table 13 and Fig. 4 yields three key empirical insights:

(1) **Optimal Spatial Window:** As illustrated in Fig. 4(a), Configuration Rank 1 (11 × 11, batch size 32) achieves the best trade-off, attaining 98.05% OA and 97.78 Kappa in 8.80 minutes. Although smaller patches (9 × 9, batch 32) train in 6.00 minutes, OA drops to 95.79% because the localized window fails to encapsulate spatial texture boundaries. Conversely, expanding patch size to 15 × 15 and 17 × 17 inflates training latency to 16.60 and

turnitin Page 32 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 33 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

Table 16. Performance Degradation Analysis Under Data Scarcity Stress Testing Across Three Benchmarks (Patch 11 × 11, Batch 32, 200 Epochs).

Dataset Benchmark	1% Train Ratio	2% Train Ratio	5% Train Ratio	10% Train Ratio	Degradation (10% → 1%)
Indian Pines	70.12%	81.55%	93.91%	97.84%	-27.73%
University of Pavia	95.41%	98.04%	99.51%	99.83%	-4.43%
Salinas Valley	97.73%	99.11%	99.79%	99.93%	-2.20%

19.40 minutes without accuracy gains (97.82% and 97.75% OA). Because Indian Pines has a 20-meter ground sampling distance, a 17 × 17 window spans 340 × 340 m^2, crossing agricultural boundaries and introducing spectral contamination from neighboring fields.

(2) **Batch Size Dynamics:** Across all patch dimensions, smaller batch sizes ($B = 32$ and $B = 64$) systematically outperform larger batch sizes ($B = 128$ and $B = 256$). For instance, with an 11 × 11 patch, scaling B from 32 to 256 degrades OA from 98.05% to 94.62% (-3.43%). Under severe class imbalance, large batches cause gradient steps to be overwhelmingly dominated by frequent majority classes (e.g., Soybean-mintill with 2,468 pixels), washing out minority gradient contributions. A smaller batch size of 32 provides higher stochastic gradient variance, enabling the optimizer to escape sharp local minima and preserve minority decision boundaries.

(3) **Cross-Dataset Consistency:** On University of Pavia (Fig. 4(b)) and Salinas Valley (Fig. 4(c)), the 11 × 11 / batch 32 setup maintains competitive performance: 99.82% OA on Pavia (Rank 6, 12.9 min vs. Rank 1 at 13 × 13 with 99.86% OA in 25.6 min) and 99.92% OA on Salinas Valley (Rank 11, 57.7 min vs. Rank 1 at 17 × 17 with 100.00% OA in 116.8 min). Adopting 11 × 11 with batch size 32 universally cuts training latency by half while sacrificing less than 0.08% in Overall Accuracy.

5.4 Data Scarcity Stress Testing and Degradation Dynamics

Operational deployment in remote sensing frequently encounters label scarcity, where field surveys provide ground truth for only a tiny fraction of pixels. To test the generalization limit of SGSCF-Net under extreme data starvation, we conduct twelve standardized stress-tests across training ratios of 1%, 2%, 5%, and 10% per class while maintaining constant architecture hyperparameters (11 × 11 patch, batch size 32, 200 epochs).

Fig. 5 illustrates the empirical performance degradation trajectories across the three benchmarks, while Table 16 reports the exact numerical metrics and percentage degradation.

Table 16 and Fig. 5 expose distinct, dataset-specific sensitivity behaviors:

- **Salinas Valley (-2.20% drop):** As visualized in Fig. 5, Salinas Valley exhibits exceptional resilience, retaining 97.73% OA under a 1% training ratio. Due to large contiguous fields and high intra-class spatial homogeneity, the SBAM and SCFG modules extract representative class descriptors even from minimal training patches.
- **University of Pavia (-4.43% drop):** Despite high intra-class spectral variance among urban materials, Pavia retains 95.41% OA at 1% training data. CBSA's complementary branch gating preserves distinct spectral signatures (such as Asphalt versus Bitumen) without confusing spatial edge transitions.
- **Indian Pines (-27.73% drop):** Indian Pines experiences a steep performance decline from 97.84% at 10% down to 70.12% at 1%. As shown in Fig. 5, a sharp degradation knee occurs below 5% training ratio. This decline reflects a mathematical sample scarcity floor rather than an architectural defect, consistent with effective sample volume theory [8] where representation quality degrades sharply when effective sample counts drop below class manifold rank.

turnitin Page 33 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 34 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359



Fig. 5. Performance degradation dynamics of SGSCF-Net under data scarcity stress testing across training ratios of 1%, 2%, 5%, and 10% on Indian Pines, University of Pavia, and Salinas Valley.

Table 17. Class-Level Performance Extremes Under Extreme 1% Training Scarcity.

Dataset	Best Performing Class	Worst Performing Class
Indian Pines	Hay-windrowed (99.27%)	Oats (7.66%)
University of Pavia	Painted metal sheets (99.62%)	Self-Blocking Bricks (84.70%)
Salinas Valley	Brocoli_green_weeds_2 (100.00%)	Fallow (89.47%)

Table 17 details the per-dataset class extremes under 1% data. On Indian Pines, the Oats category contains only 20 total labeled pixels across the entire scene. At a 1% sampling ratio, the training set receives only $\max(\lfloor 0.01 \times 20 \rfloor, 3)$ or fewer than 2 distinct training patches. With only 1 or 2 patches available, learning high-dimensional 3D spatial-spectral features becomes mathematically indeterminate, dropping Oats accuracy to 7.66%. Conversely, well-sampled categories such as Hay-windrowed maintain 99.27% accuracy. Crucially, while baseline models such as SSAN collapse into predicting majority categories exclusively under this extreme regime, SGSCF-Net's minority-aware hybrid loss ($\mathcal{L}_{\text{Total}} = 0.7\mathcal{L}_{\text{Spread}} + 0.3\mathcal{L}_{\text{Focal}}$) prevents complete boundary collapse, preserving stable 70.12% overall accuracy across the remaining fifteen categories, corroborating minority stabilization insights from prior imbalanced HSI literature [8, 19, 42].

6 Conclusion and Future Work

In this article, we addressed the structural dilemmas of hyperspectral image classification—namely, atmospheric absorption noise, spatial-spectral entanglement, quadratic computational complexity, and minority class boundary

turnitin Page 34 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 35 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

collapse. We proposed the **Spectral-Guided Spatial Cross-Attention Fusion Network (SGSCF-Net)**, an end-to-end deep learning framework that integrates three targeted architectural modules:

(1) **Spectral Band Attention Module (SBAM):** Operates directly on raw input bands before convolution, dynamically attenuating noisy atmospheric absorption channels early in the feature pipeline.
(2) **Complementary Branch Spectral Attention (CBSA):** Employs lightweight grouped 1D convolutions to enforce strictly complementary gating (α versus $1-\alpha$) between spectral-only and spatial-spectral 3D convolutional streams, guaranteeing feature specialization without parameter overhead.
(3) **Second-Order Covariance Fusion Gate (SCFG):** Replaces conventional first-order spatial pooling with bilinear outer-product covariance pooling ($G = f_{\text{ch}} \otimes f_{\text{sp}}$), capturing rich spatial-channel co-activation statistics while introducing only 82 learnable parameters.
(4) **Minority-Aware Hybrid Loss:** Combines dynamic margin-scheduled Spread Loss and inverse-frequency class-weighted Focal Loss to establish wide decision margins around hard boundary samples, mitigating class collapse under severe sample scarcity.

Extensive empirical evaluations spanning 120+ experimental runs and eight standardized baseline models demonstrate the effectiveness of SGSCF-Net. On the challenging Indian Pines benchmark, SGSCF-Net achieves 98.05% Overall Accuracy, 94.82% Average Accuracy, and 97.78 Kappa within 8.80 minutes of training, outperforming standard Vision Transformers (+5.35% OA) and cutting training latency by 52.2% relative to A2S2K-ResNet and 76.7% relative to HyperRoPE-HiT. Furthermore, SGSCF-Net attains 99.82% OA on University of Pavia and 99.92% OA on Salinas Valley. The complete network comprises only 308,097 trainable parameters (~1.2 MB checkpoint), eliminating the quadratic memory constraints of Transformer self-attention.

Despite these advantages, several challenges remain. While SGSCF-Net operates with linear computational complexity relative to spatial token sequences, 3D convolutions still demand moderate tensor memory during volumetric sliding-window feature extraction. In future work, we will investigate:

- **Hardware-Aware Edge Quantization:** Developing 8-bit integer (INT8) and FP8 quantization pipelines to enable real-time onboard inference on field-programmable gate arrays (FPGAs) and edge neural processing units (NPUs) aboard UAVs and CubeSats.
- **Self-Supervised Spectral Pre-Training:** Incorporating masked autoencoding on large uncurated hyperspectral planetary cubes to pre-train SBAM and CBSA representations, elevating few-shot discrimination on extreme minority classes (such as Oats) where ground-truth labels are limited.
- **Adaptive Deformable Kernel Topology:** Investigating spatially deformable 3D kernels within the CBSA branch to dynamically adjust receptive field boundaries along irregular land-cover transitions.
- **Selective State-Space Sequence Modeling:** Exploring linear-time selective state-space layers [16, 30] within the post-SCFG spatial path to further optimize long-range contextual receptive fields without incurring quadratic self-attention latency.

Acknowledgments

This research was conducted at the National Institute of Technology (NIT) Raipur. The authors express their gratitude to the NASA Jet Propulsion Laboratory and the University of Pavia for providing the open-access hyperspectral benchmarks.

turnitin Page 35 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 36 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

References

[1] Dalal Al-Alimi, Zhihua Cai, Mohammed A. A. Al-qaness, Abdelghani Dahou, Eman Ahmed Alawamy, and Sakinatu Issaka. 2022. Compression and reinforce variation with convolutional neural networks for hyperspectral image classification. *Applied Soft Computing* 131 (2022), 109650.

[2] Amina Ben Hamida, Alexandre Benoit, Patrick Lambert, and Christian Ben Amar. 2018. Deep Learning for Hyperspectral Image Classification: An Overview Based on 3-D Convolutional Neural Networks. *IEEE Geoscience and Remote Sensing Magazine* 6, 2 (2018), 42–63.

[3] José M. Bioucas-Dias, Antonio Plaza, Gustavo Camps-Valls, Paul Scheunders, Nasser Nasrabadi, and Jocelyn Chanussot. 2013. Hyperspectral Remote Sensing Data Analysis and Future Challenges. *IEEE Geoscience and Remote Sensing Magazine* 1, 2 (2013), 6–36.

[4] Gustavo Camps-Valls and Lorenzo Bruzzone. 2005. Kernel-based methods for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing* 43, 6 (2005), 1351–1362.

[5] Kaidi Cao, Colin Wei, Adrien Gaidon, Nikos Aréchiga, and Tengyu Ma. 2019. Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32. 1567–1578.

[6] Yushi Chen, Hao Jiang, Chenghao Li, Xiuping Jia, and Pedram Ghamisi. 2016. Deep Feature Extraction and Classification of Hyperspectral Images Based on Convolutional Neural Networks. *IEEE Transactions on Geoscience and Remote Sensing* 54, 10 (2016), 6232–6251.

[7] Yushi Chen, Zhouhan Lin, Xing Zhao, Gang Wang, and Yanfeng Gu. 2014. Deep Learning-Based Classification of Hyperspectral Data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 7, 6 (2014), 2094–2107.

[8] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. 2019. Class-Balanced Loss Based on Effective Number of Samples. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9268–9277.

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*.

[10] Leyuan Fang, Ning He, Shutao Li, Antonio J. Plaza, and Javier Plaza. 2018. A New Spatial–Spectral Feature Extraction Method for Hyperspectral Images Using Local Covariance Matrix Representation. *IEEE Transactions on Geoscience and Remote Sensing* 56, 6 (2018), 3534–3546.

[11] Mathieu Fauvel, Jón Atli Benediktsson, Jocelyn Chanussot, and Johannes R. Sveinsson. 2008. Spectral and Spatial Classification of Hyperspectral Data Using SVMs and Morphological Profiles. *IEEE Transactions on Geoscience and Remote Sensing* 46, 11 (2008), 3804–3814.

[12] Mathieu Fauvel, Yuliya Tarabalka, Jón Atli Benediktsson, Jocelyn Chanussot, and James C. Tilton. 2013. Advances in Spectral-Spatial Classification of Hyperspectral Images. *Proc. IEEE* 101, 3 (2013), 652–675.

[13] Zilin Gao, Jiangtao Xie, Qilong Wang, and Peihua Li. 2019. Global Second-Order Pooling Convolutional Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 3024–3033.

[14] Pedram Ghamisi, Javier Plaza, Yushi Chen, Jun Li, and Antonio J. Plaza. 2017. Advanced Spectral Classifiers for Hyperspectral Images: A Review. *IEEE Geoscience and Remote Sensing Magazine* 5, 1 (2017), 8–32.

[15] Robert O. Green, Michael L. Eastwood, Charles M. Sarture, Thomas G. Chrien, Mikael Aronsson, Bruce J. Chippendale, Jessica A. Faust, Betina E. Pavri, Christopher J. Chovit, Manuel Solis, Martin R. Olah, and Orlesa Williams. 1998. Imaging Spectroscopy and the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). *Remote Sensing of Environment* 65, 3 (1998), 227–248.

[16] Albert Gu and Tri Dao. 2023. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv preprint arXiv:2312.00752* (2023).

[17] Wenhui Guo, Hailiang Ye, and Feilong Cao. 2022. Feature-Grouped Network With Spectral–Spatial Connected Attention for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–13.

[18] Renlong Hang, Zhu Li, Qingshan Liu, Pedram Ghamisi, and Shuvra S. Bhattacharyya. 2021. Hyperspectral Image Classification With Attention-Aided CNNs. *IEEE Transactions on Geoscience and Remote Sensing* 59, 3 (2021), 2281–2293.

[19] Juan M. Haut, Mercedes E. Paoletti, Javier Plaza, Antonio Plaza, and Jun Li. 2022. Generative Adversarial Minority Oversampling for Spectral–Spatial Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–12.

[20] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. (2016), 770–778.

[21] Mingyi He, Bo Li, and Huahui Chen. 2018. Multi-scale 3D deep convolutional neural network for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing* 56, 9 (2018), 5262–5273.

[22] Na He, Mercedes E. Paoletti, Juan M. Haut, Shutao Li, Antonio Plaza, and Javier Plaza. 2019. Feature Extraction With Multiscale Covariance Maps for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 57, 2 (2019), 755–769.

[23] Xixi He, Yushi Chen, and Zhouhan Lin. 2021. Spatial-Spectral Transformer for Hyperspectral Image Classification. *Remote Sensing* 13, 3 (2021), 498.

[24] Danfeng Hong, Zheng Han, Jing Yao, Lianru Gao, Bing Zhang, Antonio Plaza, and Jocelyn Chanussot. 2022. SpectralFormer: Rethinking Hyperspectral Image Classification With Transformers. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–15.

[25] Jie Hu, Li Shen, and Gang Sun. 2018. Squeeze-and-Excitation Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 7132–7141.

[26] Wei Hu, Yangyu Huang, Li Wei, Fan Zhang, and Hengchao Li. 2015. Deep Convolutional Neural Networks for Hyperspectral Image Classification. *Journal of Sensors* 2015 (2015), 1–12.

[27] Zirak Khan, Noyon Dey, Kavya Kathiravan, Seung-Chul Yoon, and Suchendra Bhandarkar. 2025. Spatial-Spectral Transformer With Patch-Local Mixed-Axis 2D Rotary Position Embedding for Hyperspectral Image Classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 18 (2025), 4412–4425.

turnitin Page 36 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 37 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

[28] Peihua Li, Jiangtao Xie, Qilong Wang, and Zilin Gao. 2018. Towards Faster Training of Global Covariance Pooling Networks by Iterative Matrix Square Root Normalization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 947–955.

[29] Rui Li, Siyuan Zheng, Chenxi Duan, Yongjun Zheng, and Lu Su. 2020. Double-Branch Dual-Attention Mechanism Network for Hyperspectral Image Classification. *Remote Sensing* 12, 8 (2020), 1307.

[30] Xiang Li, Yanan Ding, Wei Li, Lingling Ran, and Meng Zhao. 2024. MambaHSI: Spatial-Spectral Mamba for Hyperspectral Image Classification. *IEEE Geoscience and Remote Sensing Letters* 21 (2024), 1–5.

[31] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal Loss for Dense Object Detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2980–2988.

[32] Tsung-Yi Lin, Aruni RoyChowdhury, and Subhransu Maji. 2015. Bilinear CNN Models for Fine-Grained Visual Recognition. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 1449–1457.

[33] Farid Melgani and Lorenzo Bruzzone. 2004. Classification of hyperspectral remote sensing images with support vector machines. *IEEE Transactions on Geoscience and Remote Sensing* 42, 8 (2004), 1778–1790.

[34] Lichao Mou, Sudipan Saha, Yuansheng Hua, Francesca Bovolo, Lorenzo Bruzzone, and Xiao Xiang Zhu. 2022. Deep Reinforcement Learning for Band Selection in Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–14.

[35] Mercedes E. Paoletti, Juan M. Haut, Javier Plaza, and Antonio Plaza. 2018. Capsule Networks for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 57, 4 (2018), 2114–2131.

[36] Antonio Plaza, Jón Atli Benediktsson, Joseph W. Boardman, Jason Brazile, Lorenzo Bruzzone, Gustavo Camps-Valls, Jocelyn Chanussot, Mathieu Fauvel, Paolo Gamba, Anthony Gualtieri, Mattia Marconcini, James C. Tilton, and Giovanna Trianni. 2009. Recent Advances in Techniques for Hyperspectral Image Processing. *Remote Sensing of Environment* 113 (2009), S110–S122.

[37] Yuhao Qing, Wenyi Liu, Liuyan Feng, and Wanjia Gao. 2021. Improved Transformer Net for Hyperspectral Image Classification. *Remote Sensing* 13, 11 (2021), 2216.

[38] Swalpa Kumar Roy, Ankur Deria, Chiranjibi Shah, Juan M. Haut, Qian Du, and Antonio Plaza. 2023. Spectral–Spatial Morphological Attention Transformer for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–15.

[39] Swalpa Kumar Roy, Gopal Krishna, Shiv Ram Dubey, and Bidyut B. Chaudhuri. 2020. HybridSN: Exploring 3-D-2-D CNN Feature Hierarchy for Hyperspectral Image Classification. *IEEE Geoscience and Remote Sensing Letters* 17, 2 (2020), 277–281.

[40] Swalpa Kumar Roy, Somnath Manna, Tao Song, and Lorenzo Bruzzone. 2020. Attention-Based Adaptive Spectral-Spatial Kernel ResNet for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 59, 9 (2020), 7831–7843.

[41] Sara Sabour, Nicholas Frosst, and Geoffrey E. Hinton. 2018. Matrix Capsules with EM Routing. In *International Conference on Learning Representations (ICLR)*.

[42] Yi Su, Xiaojun Li, Junping Yao, Chengrong Dong, and Yao Wang. 2023. A Spectral–Spatial Feature Rotation-Based Ensemble Method for Imbalanced Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–15.

[43] Heng Sun, Xiangtao Zheng, Xiaoqiang Lu, and Shengyang Wu. 2019. Spectral-Spatial Attention Network for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 58, 5 (2019), 3232–3245.

[44] Le Sun, Guangrui Zhao, Yuhui Zheng, and Zebin Wu. 2022. Spectral-Spatial Feature Tokenization Transformer for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–14.

[45] Lu Wang, Shaohui Mei, Qian Du, and Chen Chen. 2023. CSCANet: Class-Specific Cross-Attention Network for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–15.

[46] Qilong Wang, Banggu Wu, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. 2020. ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 11534–11542.

[47] Xinyu Wang, Kun Tan, Peijun Du, and Yuting Chen. 2021. Deep Matrix Capsules With EM Routing for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 59, 12 (2021), 10368–10382.

[48] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon. 2018. CBAM: Convolutional Block Attention Module. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 3–19.

[49] Zhaohui Xue, Mengxue Zhang, Yifeng Liu, and Peijun Du. 2021. Attention-Based Second-Order Pooling Network for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 59, 11 (2021), 9600–9615.

[50] Xiaofei Yang, Weijia Cao, Yao Lu, and Yicong Zhou. 2022. Hyperspectral Image Transformer Classification Networks. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–15.

[51] Mengmeng Zhang, Wei Li, and Qian Du. 2018. Diverse Region-Based CNN for Hyperspectral Image Classification. *IEEE Transactions on Image Processing* 27, 6 (2018), 2623–2634.

[52] Wenzhi Zhao and Shihong Du. 2016. Spectral–Spatial Feature Extraction for Hyperspectral Image Classification: A Dimension Reduction and Deep Learning Approach. *IEEE Transactions on Geoscience and Remote Sensing* 54, 8 (2016), 4544–4554.

[53] Jianwei Zheng, Yuchao Feng, Cong Bai, and Jinglin Zhang. 2021. Hyperspectral Image Classification Using Mixed Convolutions and Covariance Pooling. *IEEE Transactions on Geoscience and Remote Sensing* 59, 1 (2021), 522–534.

[54] Zilong Zhong, Jonathan Li, Zhiming Luo, and Michael Chapman. 2017. Spectral-Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework. *IEEE Transactions on Geoscience and Remote Sensing* 56, 2 (2017), 847–858.

turnitin Page 37 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

turnitin Page 38 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359

[55] Minghao Zhu, Licheng Jiao, Fang Liu, Shuyuan Yang, and Jianing Wang. 2021. Residual Spectral–Spatial Attention Network for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 59, 1 (2021), 449–462.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

turnitin Page 38 of 38 - Integrity Submission Submission ID trn:oid:::3618:156538359